Summary Report, v3: Aggregation Views from the Literature, Every Combination, and Parameters with Standard Errors

2026

1 Setup

1.1 Shared setup (identical across versions)

- **Suite.** 31 datasets: 11 entity matching (EM), 6 schema matching (SM), 11 table question answering and understanding (TQA), 3 financial documents (FIN). Official test or dev splits only; 150 items per dataset by seeded stratified sampling, 100 flagged for answering (OfficeQA Pro V2: 90; Alaska schema: 93); 3,083 answered items per model. Entity-matching pools are 50/50 match / non-match (Papadakis–Christen keeps its real 1.8 % ratio); schema matching is one question per source column with a candidate list; BIRD is graded on the executed result of the gold SQL.
- **Answers.** Ten models (Claude Opus 5, Claude Sonnet 5, GPT-5.5, GPT-5.4-mini, Gemini 3.8 Flash, Grok 4, DeepSeek-chat, DeepSeek-R1, gpt-oss-120b, Llama 3.3 70B); one prompt per item ending in `FINAL ANSWER: <answer>`; temperature 0 where accepted (Claude 5 and GPT-5.5 ran at provider defaults, recorded per row); output budget 700 tokens with mechanical re-asks at 2,500 and 8,000 for replies cut off before a final line; every answer cached with tokens, cost and finish reason. The same cache feeds every version, so leaderboards are directly comparable.
- **Scoring primitives.** Extraction of the final line, Unicode / case / punctuation normalisation, number parsing with sign, percent and scale words, alias lists per gold (best alias wins), list golds, a hedge flag. Native metric per dataset: numeric tolerance (1 %) for numeric golds, exact match for text and boolean, ROUGE-L for free-form.
- **Coverage rule.** A model is in the study only if it covered ≥ 90 % of the suite; a dataset counts only items every model answered and only with ≥ 20 such items.
- **Leaderboard machinery.** Bradley–Terry strengths from item-level pairwise wins (MLE, ridge 0.01, SE from the observed information) as the headline pooling; Borda, Kemeny (with cost), Copeland, mean-z, reciprocal-rank fusion and family-balanced z as checks; 1,000 item bootstraps of the Borda and Bradley–Terry ranks; Friedman with Iman–Davenport, Nemenyi CD, Holm-corrected pairwise Wilcoxon across datasets; paired permutation and Wilcoxon per item within datasets (Holm); Kendall's W among datasets; majority-class baselines for boolean sets; post-stratified official-split estimates with a weighted bootstrap and positive-class F1 at the official match ratio; floor / ceiling / self-stability diagnostics per dataset.
- **Sampled designs.** Uniform-lattice, Dirichlet-snapped and Latin-hypercube samples on the weight simplex at step 0.05 (20 seeds each) with centred L_2 discrepancy and nearest-neighbour distances, compared with the exhaustive optimum where one exists.
- **Reporting.** Each version's `run_all.sh` regenerates its `REPORT.md` from the cached answers; every number in this document is read from those output files.

1.2 What v3 searches

- **Per-answer rule.** None is searched: each dataset keeps its own native metric (numeric tolerance, exact match or ROUGE-L). v3 asks a different question from v2 and v4: not "how should an answer be scored" but "how should per-dataset scores be pooled into one ranking".
- **Eight aggregation views** from the literature, each mapping the models × datasets × items score tensor to one number per model: mean raw score; baseline-normalised mean (Open LLM Leaderboard v2: (raw − random baseline)/(1 − baseline), clipped at 0); mean z across datasets; mean win rate (HELM: fraction of other models beaten per dataset); Borda; Kemeny score; Bradley–Terry strength from item-level pairwise wins (Chatbot Arena); Rasch ability from a one-parameter IRT fit with items as difficulties (tinyBenchmarks). Views are z-scored across models before mixing.
- **Candidates.** Every non-empty equal-weight subset of the eight views (255, including the eight corners and the uniform mix), plus 100 uniform-lattice and 100 Latin-hypercube mixes at step 0.05: 455 candidates.
- **Objective.** $J = 0.5\,\text{stability} + 0.25\,\text{transitivity} + 0.25\,\text{reference}$: stability = mean Kendall τ between the mixed ranking on item-bootstrap resamples and on the full data; transitivity = τ to the Copeland ranking of the per-dataset scores; reference = τ to the Kemeny consensus of the eight single-view rankings. J measures how reproducible and internally consistent a pooled ranking is, not whether it is right.
- **Parameters saved.** Bradley–Terry strengths with analytic SE (vs reference) and bootstrap SD, Rasch abilities with SE and item difficulties with SE, Kemeny order and cost, Copeland, Borda, the win matrix, and per-bootstrap-draw parameters (100 item resamples). Every combination's member parameters, mixed scores, ranks and rank intervals are stored under `data/combos/`.
- **Honesty.** Split-half: 20 random half-splits of the items; the best candidate on half A (20 inner resamples) is scored on half B and compared with pure Bradley–Terry, the uniform mix and the full-data best on half B.
- **Leaderboards.** Native metric (headline) and exact match, each pooled by Bradley–Terry with the full significance battery; the five best mixes are reported with bootstrap rank intervals.

2 Results

2.1 Single views and their mixes

Every view ranks the same model last and nearly the same three models first; the views differ in the middle (Figure 1). Used alone, the rank-based and mean-based views ($J \approx 0.93$) beat the two fitted-parameter views (Bradley–Terry 0.891, Rasch 0.904) because the fitted views move more under item resampling. Mixing helps a little: the uniform mix reaches 0.941 and the best four-view subset 0.948 (Figure 3, Table 25). The whole candidate range is 0.882–0.948; the sampled designs land within 0.002 of the best subset (Figure **??**).

Table 1: v3 audit summary.

Single views	J alone from 0.886 (baseline-normalised mean) to 0.932 (Borda); Bradley–Terry alone 0.891, Rasch alone 0.904
Best mix	subset_10011001: mean raw 0.25, mean win rate 0.25, Borda 0.25, Rasch ability 0.25; $J = 0.948$ (stability 0.923, transitivity 0.94, reference 1.00); uniform mix $J = 0.941$
Candidates	455 (255 subsets + 100 lattice + 100 LHS); J range 0.882–0.948; best per design: subset 0.948, lattice 0.946, LHS 0.946
Split-half honesty	selected mix J 0.935 on half A, 0.903 on half B (optimism gap 0.032); on half B pure BT 0.879, uniform 0.917, full-data best 0.913
Selected minus pure BT (held out)	+0.024, 95 % CI [−0.034, +0.111]; selected ahead in 70 % of splits; 10 distinct mixes selected in 20 splits
Parameters	BT strengths span -0.161 to +0.063 (bootstrap SD ≈ 0.005); Rasch abilities -1.77 to +0.74 (SE ≈ 0.10); Kemeny cost 329 of 1395
Leaderboard (native, BT)	first Gemini 3.8 Flash, then GPT-5.5, Claude Opus 5; last Llama 3.3 70B; Friedman $p = 1.2\text{e-}13$; Kendall's $W = 0.27$
Leaderboard (exact match, BT)	first Claude Opus 5; τ between the native and exact-match BT rankings 0.78

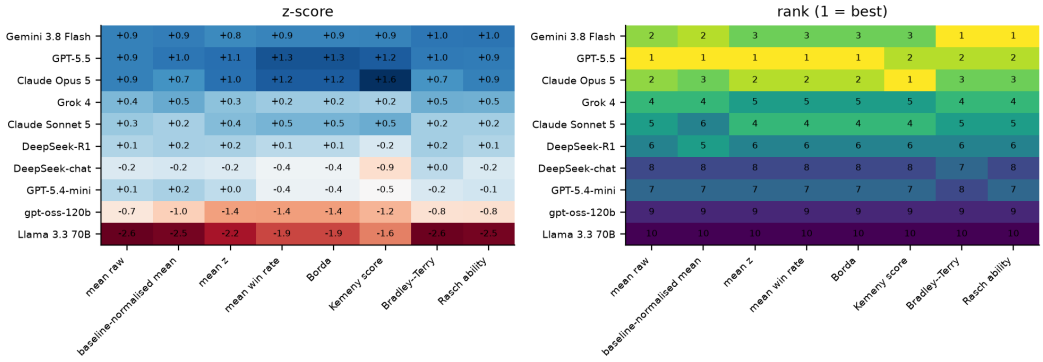


Figure 1: Each view's z-score (left) and rank (right) per model on native-metric scores.

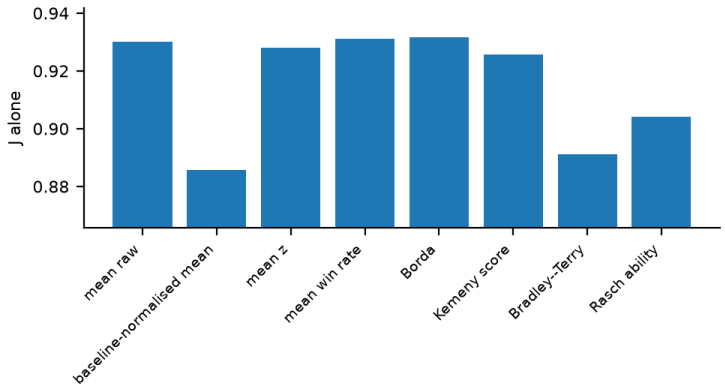


Figure 2: Objective J of each view used alone.

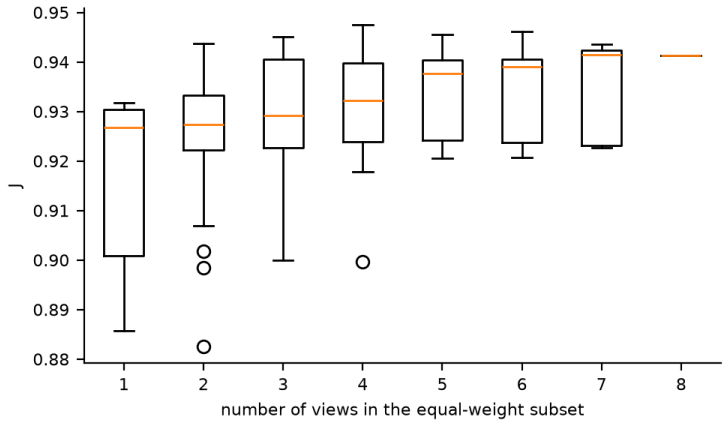


Figure 3: Objective J of the 255 equal-weight subsets by subset size.

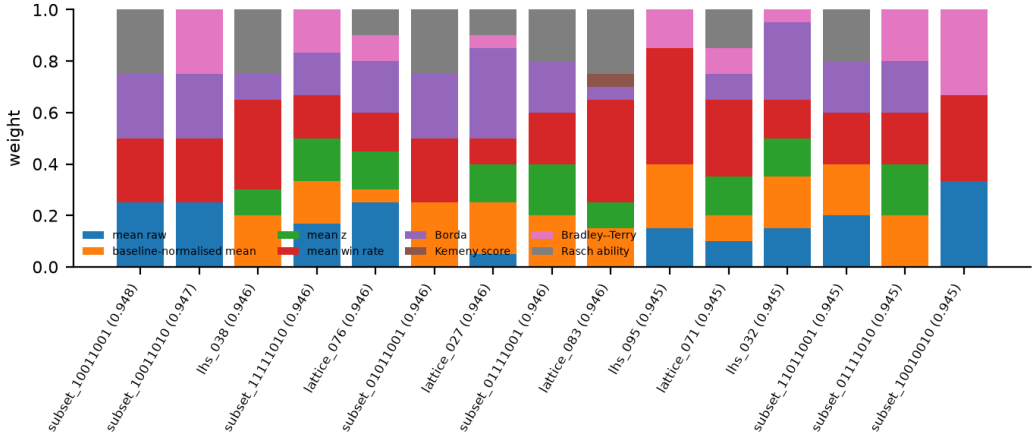


Figure 4: View weights of the fifteen best candidate mixes (label and J on the axis).

Table 2: The fifteen best candidate mixes with their weights and objective terms.

rank	label	n_views	w_mean_raw	w_baseline_norm_mean	w_z_mean	w_mean_win_rate	w_borda	w_kemeny_score	w_bradley_terry	w_irt_ability	stability	transitivity	reference	J
1	subset_10011001	4	0.250	0.000	0.000	0.250	0.250	0.000	0.000	0.250	0.923	0.944	1.000	0.948
2	subset_10011010	4	0.250	0.000	0.000	0.250	0.250	0.000	0.250	0.000	0.923	0.944	1.000	0.947
3	lhs_038	5	0.000	0.200	0.100	0.350	0.100	0.000	0.000	0.250	0.920	0.944	1.000	0.946
4	subset_11111010	6	0.167	0.167	0.167	0.167	0.167	0.000	0.167	0.000	0.920	0.944	1.000	0.946
5	lattice_076	7	0.250	0.050	0.150	0.150	0.200	0.000	0.100	0.100	0.920	0.944	1.000	0.946
6	subset_01011001	4	0.000	0.250	0.000	0.250	0.250	0.000	0.000	0.250	0.920	0.944	1.000	0.946
7	lattice_027	7	0.050	0.200	0.150	0.100	0.350	0.000	0.050	0.100	0.920	0.944	1.000	0.946
8	subset_01111001	5	0.000	0.200	0.200	0.200	0.200	0.000	0.000	0.200	0.919	0.944	1.000	0.946
9	lattice_083	6	0.000	0.150	0.100	0.400	0.050	0.050	0.000	0.250	0.919	0.944	1.000	0.946
12	lhs_095	4	0.150	0.250	0.000	0.450	0.000	0.000	0.150	0.000	0.919	0.944	1.000	0.945
11	lattice_071	7	0.100	0.100	0.150	0.300	0.100	0.000	0.100	0.150	0.919	0.944	1.000	0.945
10	lhs_032	6	0.150	0.200	0.150	0.150	0.300	0.000	0.050	0.000	0.919	0.944	1.000	0.945
13	subset_11011001	5	0.200	0.200	0.000	0.200	0.200	0.000	0.000	0.200	0.918	0.944	1.000	0.945
14	subset_01111010	5	0.000	0.200	0.200	0.200	0.200	0.000	0.200	0.000	0.918	0.944	1.000	0.945
15	subset_10010010	3	0.333	0.000	0.000	0.333	0.000	0.000	0.333	0.000	0.918	0.944	1.000	0.945

2.2 Honest selection

The split-half test (Figure 5) is the honesty check for the choice of mix. The mix selected on half A loses 0.032 of J on half B, and 20 splits select 10 different mixes. On the held-out half the selected mix beats pure Bradley–Terry by +0.024 with an interval that includes zero, and the plain uniform mix (0.917) does as well as the selected one (0.903). Mixing views is therefore defensible but not demonstrably better than the uniform mix; pure Bradley–Terry is the least reproducible single rule on this suite.

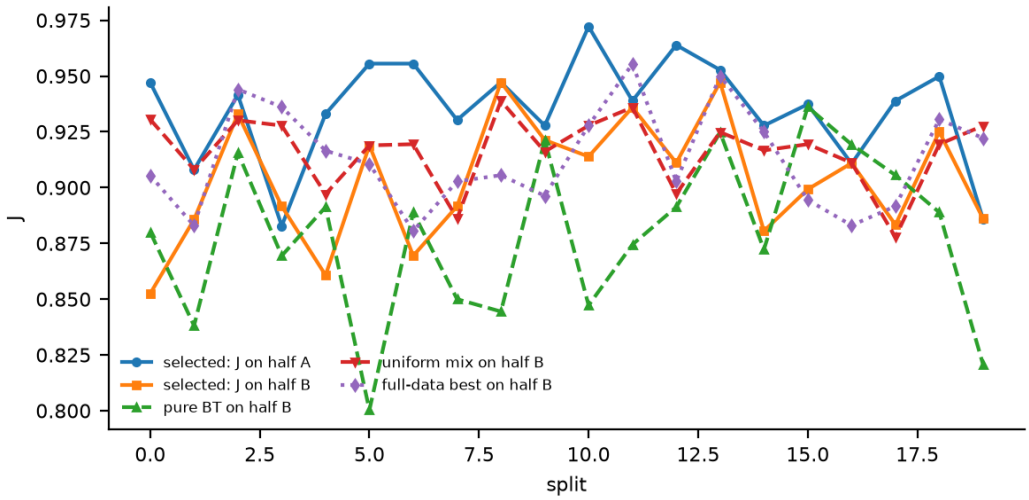


Figure 5: Split-half honesty check: the mix selected on half A scored on both halves, against pure Bradley–Terry, the uniform mix and the full-data best on half B.

Table 3: Split-half summary.

quantity	value
splits	20
inner_resamples	8
J_A_selected_mean	0.935
J_B_selected_mean	0.903
optimism_gap	0.032
J_B_pure_bt_mean	0.879
J_B_uniform_mean	0.917
J_B_full_best_mean	0.913
selected_minus_pure_bt	"mean": 0.02427983120362126, "ci95": [-0.03387181157278865, 0.11109314599900576], "share_selected_ahead": 0.7

2.3 Parameters with standard errors

Table 18 and Figure 6 give the fitted Bradley–Terry strengths and Rasch abilities. The analytic standard errors assume the model is right; the bootstrap SDs (Figure 7) do not, and with ten models they are the ones to quote. Rasch item difficulties are poorly determined (mean SE 7.4,

Figure **??**): with ten responses per item the one-parameter model cannot pin individual items down, which is why the Rasch view is the least stable.

Table 4: Fitted parameters with standard errors (native metric). Kemeny cost 329 of 1395.

model	BT strength	BT SE vs ref	BT bootstrap SD	Rasch ability	Rasch SE	Rasch bootstrap SD	Copeland	Kemeny position
Gemini 3.8 Flash	0.063	0.016	0.005	0.736	0.104	0.078	6.500	3
GPT-5.5	0.060	0.016	0.005	0.693	0.103	0.075	7.500	2
Claude Opus 5	0.045	0.000	0.005	0.651	0.103	0.086	8.500	1
Grok 4	0.031	0.016	0.005	0.345	0.099	0.067	4.500	5
Claude Sonnet 5	0.012	0.016	0.005	0.125	0.097	0.067	4.500	4
DeepSeek-R1	0.012	0.016	0.005	0.070	0.096	0.070	3.500	6
DeepSeek-chat	0.002	0.016	0.005	-0.155	0.094	0.067	1.500	8
GPT-5.4-mini	-0.011	0.016	0.005	-0.084	0.094	0.066	2.500	7
gpt-oss-120b	-0.053	0.016	0.006	-0.608	0.090	0.069	0.500	9
Llama 3.3 70B	-0.161	0.016	0.007	-1.775	0.085	0.083	-0.500	10

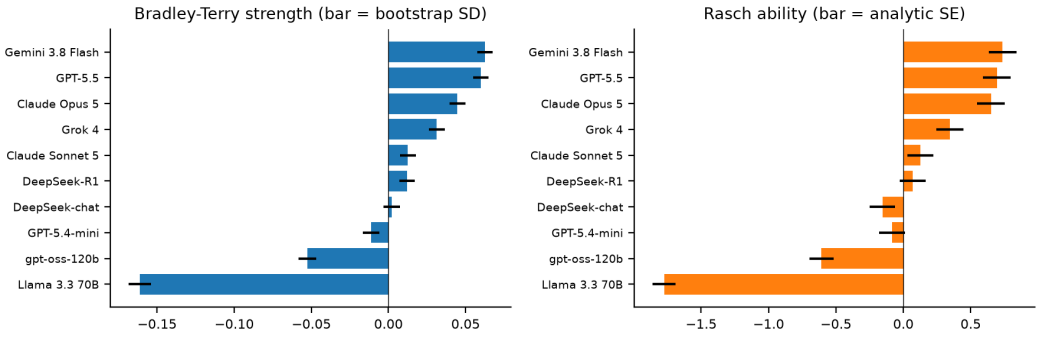


Figure 6: Fitted Bradley–Terry strengths and Rasch abilities with their uncertainty.

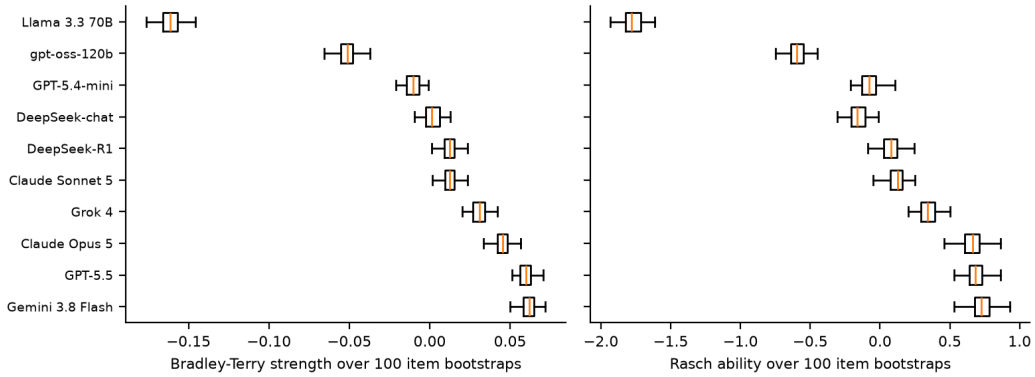


Figure 7: Bootstrap distributions of the Bradley–Terry strengths and Rasch abilities (100 item resamples).

2.4 Leaderboard

Under the native metric pooled by Bradley–Terry, Gemini 3.8 Flash is first ($P(\text{first}) = 0.62$) with GPT-5.5 and Claude Opus 5 next; Borda puts GPT-5.5 first. Exact match puts Claude Opus 5 first. The ends are fixed under every rule and pooling method; Friedman rejects equal models (p = 1.2e-13) while datasets agree only moderately ($W = 0.27$).

Table 5: Leaderboard under the headline rule (native metric).

rank	model	BT strength	SE	Borda rank	Borda 95%	P(first) BT	mean score
1	Gemini 3.8 Flash	0.063	0.016	3	[2, 4]	0.620	0.792
2	GPT-5.5	0.060	0.016	1	[1, 3]	0.370	0.793
3	Claude Opus 5	0.045	0.000	2	[1, 3]	0.000	0.792
4	Grok 4	0.031	0.016	5	[3, 6]	0.000	0.780
5	Claude Sonnet 5	0.012	0.016	4	[3, 6]	0.000	0.776
6	DeepSeek-R1	0.012	0.016	6	[4, 8]	0.000	0.772
7	DeepSeek-chat	0.002	0.016	8	[6, 8]	0.000	0.763
8	GPT-5.4-mini	-0.011	0.016	7	[6, 8]	0.000	0.771
9	gpt-oss-120b	-0.053	0.016	9	[9, 9]	0.000	0.750
10	Llama 3.3 70B	-0.161	0.016	10	[10, 10]	0.000	0.701

Table 6: Bradley–Terry rank (Borda rank in brackets) under each reported rule, Friedman p and Kendall's W among datasets.

rule	Gemini 3.8 Flash	GPT-5.5	Claude Opus 5	Grok 4	Claude Sonnet 5	DeepSeek-R1	DeepSeek-chat	GPT-5.4-mini	gpt-oss-120b	Llama 3.3 70B	Friedman p	W
native metric	1 (3)	2 (1)	3 (2)	4 (5)	5 (4)	6 (6)	7 (8)	8 (7)	9 (9)	10 (10)	1.2e-13	0.27
exact match	3 (3)	2 (2)	1 (1)	4 (5)	5 (4)	7 (6)	8 (7)	6 (8)	9 (9)	10 (10)	5.3e-14	0.28

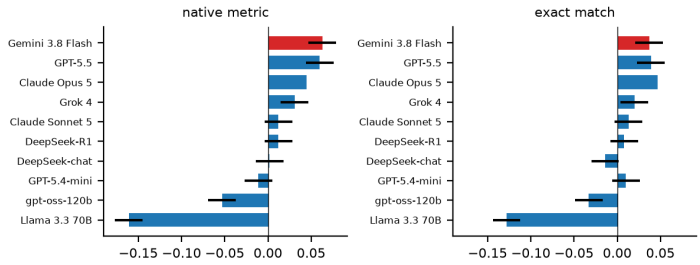


Figure 8: Bradley–Terry strengths (± SE vs reference) under every reported rule; models fixed in the headline order.

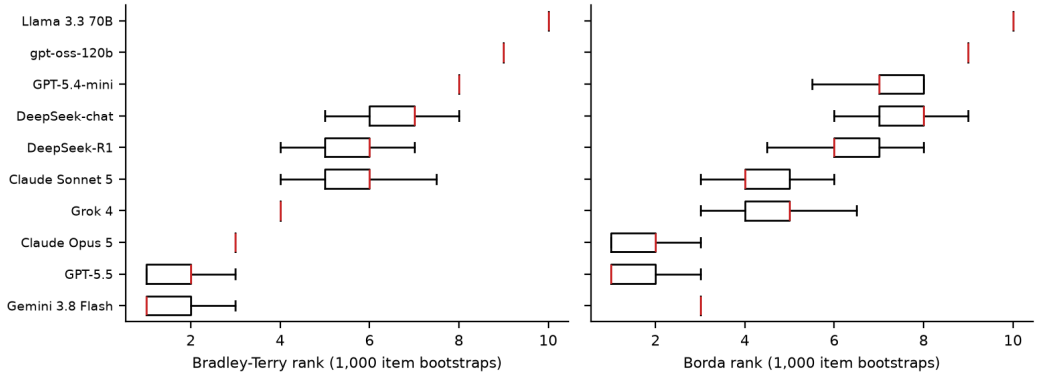


Figure 9: Bootstrap distribution of each model's rank under the headline rule (native metric).

2.5 Comparability and diagnostics

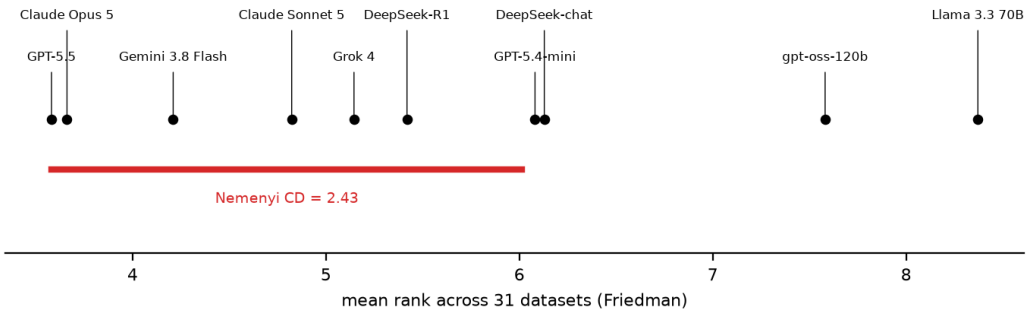


Figure 10: Critical-difference view of the Friedman test under the headline rule: mean rank per model and the Nemenyi critical difference at $\alpha = .05$.

Table 7: Official-split estimates under the headline rule averaged over the ten models.

dataset	suite_mean	official_ estimate	suite_f1_ pos	official_precision_ pos	official_recall_ pos	official_f1_ pos	official_positive_ share
abt_buy	0.970	0.986	0.968	0.926	0.949	0.936	0.108
alaska_camera	0.958	0.958	0.955	1.000	0.921	0.955	0.500
alaska_schema	0.897	0.897	–	–	–	–	–
amazon_google	0.936	0.932	0.929	0.611	0.942	0.739	0.102
bird	0.126	0.157	–	–	–	–	–
dblp_acm	0.986	0.991	0.985	0.979	0.978	0.976	0.180
dblp_scholar	0.949	0.966	0.948	0.900	0.924	0.910	0.186
fetaqa	0.468	0.468	–	–	–	–	–
fintagging	0.794	0.789	–	–	–	–	–
hitab	0.743	0.759	–	–	–	–	–
machamp	0.887	0.890	0.892	0.644	0.853	0.733	0.175
magneto_gdc	0.908	0.911	–	–	–	–	–
mmtu	0.879	0.834	–	–	–	–	–
officeqa	0.077	0.077	–	–	–	–	–
officeqa_pro_v2	0.082	0.082	–	–	–	–	–
opensanctions_ pairs	0.821	0.727	0.773	0.993	0.649	0.777	0.769
papadakis_dn	0.941	0.988	0.833	0.550	0.963	0.683	0.018
realhitbench	0.630	0.628	–	–	–	–	–
smat	0.876	0.873	–	–	–	–	–
suc	0.886	0.882	–	–	–	–	–
tab_fact	0.798	0.789	0.758	0.896	0.674	0.764	0.500
tabis	0.915	0.918	–	–	–	–	–
tablebench	0.802	0.800	–	–	–	–	–
tableeval	0.720	0.731	–	–	–	–	–
tat_qa	0.751	0.706	–	–	–	–	–
tpcdi_cells	0.343	0.375	–	–	–	–	–
valentine	0.979	0.971	–	–	–	–	–
walmart_amazon	0.958	0.975	0.947	0.832	0.927	0.875	0.094
wdc_lspc	0.982	0.983	0.983	0.965	0.979	0.971	0.264
wdc_products	0.978	0.985	0.979	0.980	0.968	0.974	0.280
wikitablequestions	0.799	0.796	–	–	–	–	–

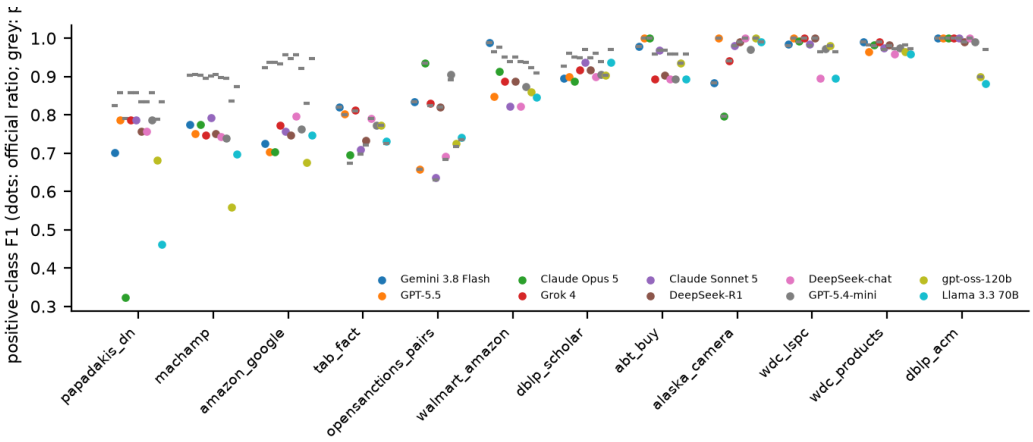


Figure 11: Positive-class F1 on the balanced pool (grey) and at the official match ratio (coloured) per model and matching dataset.

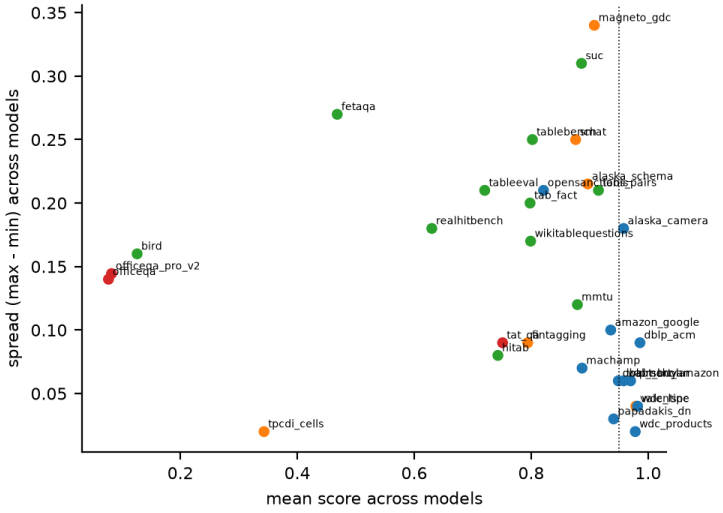


Figure 12: Dataset diagnostics: mean score against spread across models; dotted line marks the ceiling flag.

A All figures

Every figure that can be drawn from the saved outputs; source data in `data/`.

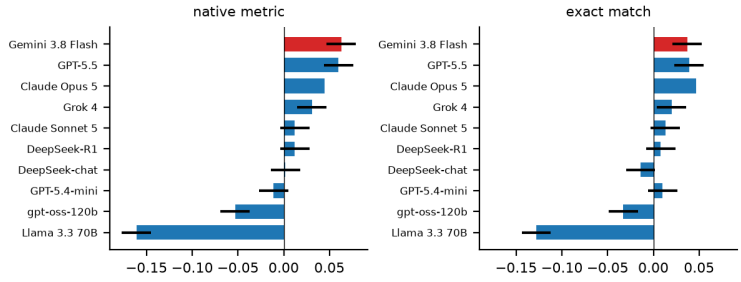


Figure 13: Bradley–Terry strengths (± SE vs reference) under every reported rule; models fixed in the headline order.

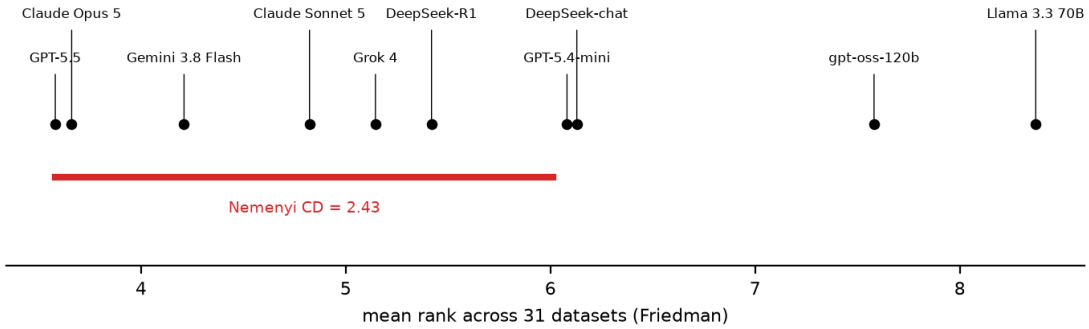


Figure 14: Critical-difference view of the Friedman test under the headline rule: mean rank per model and the Nemenyi critical difference at $\alpha = .05$.

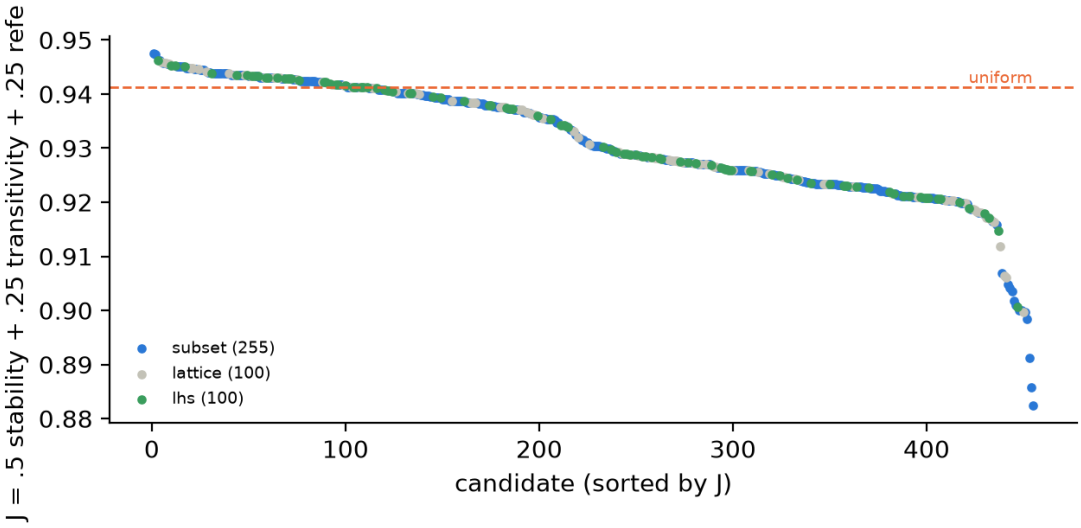


Figure 15: All 455 candidate mixes sorted by J, coloured by design.

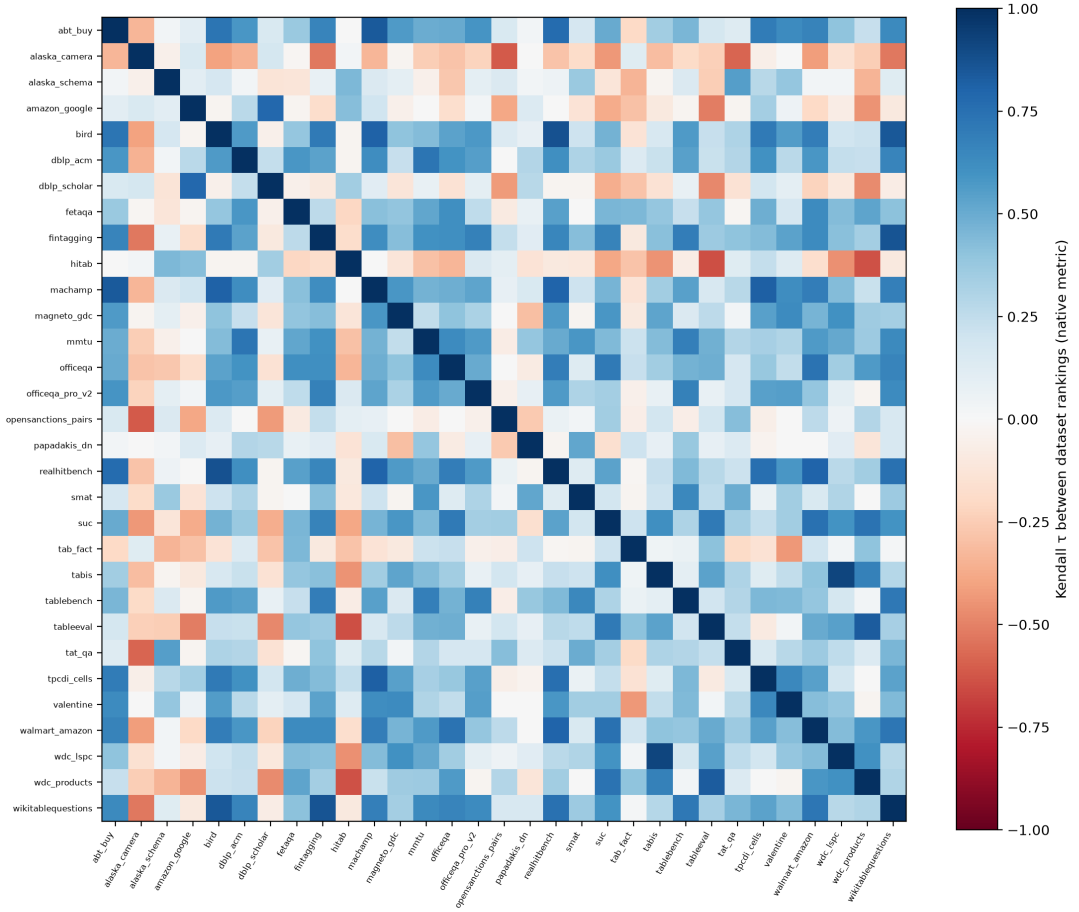


Figure 16: Kendall τ between the model rankings of each pair of datasets under the headline rule.

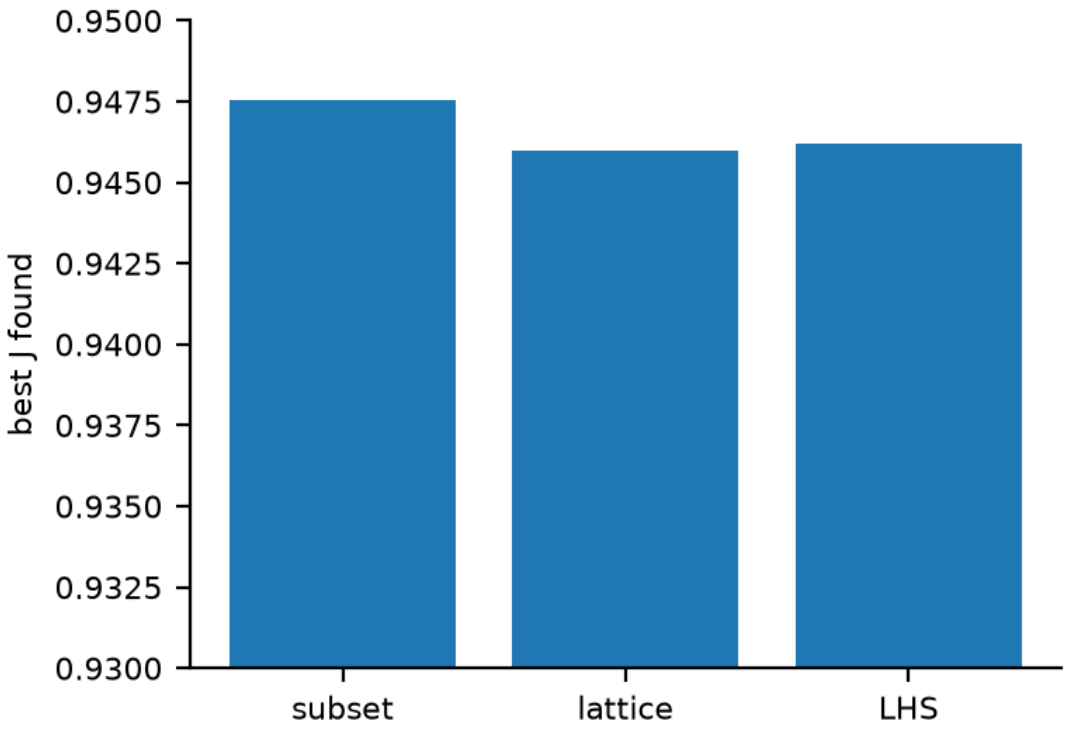


Figure 17: Best J found by the enumerated subsets and the two sampled designs of 100 mixes each.

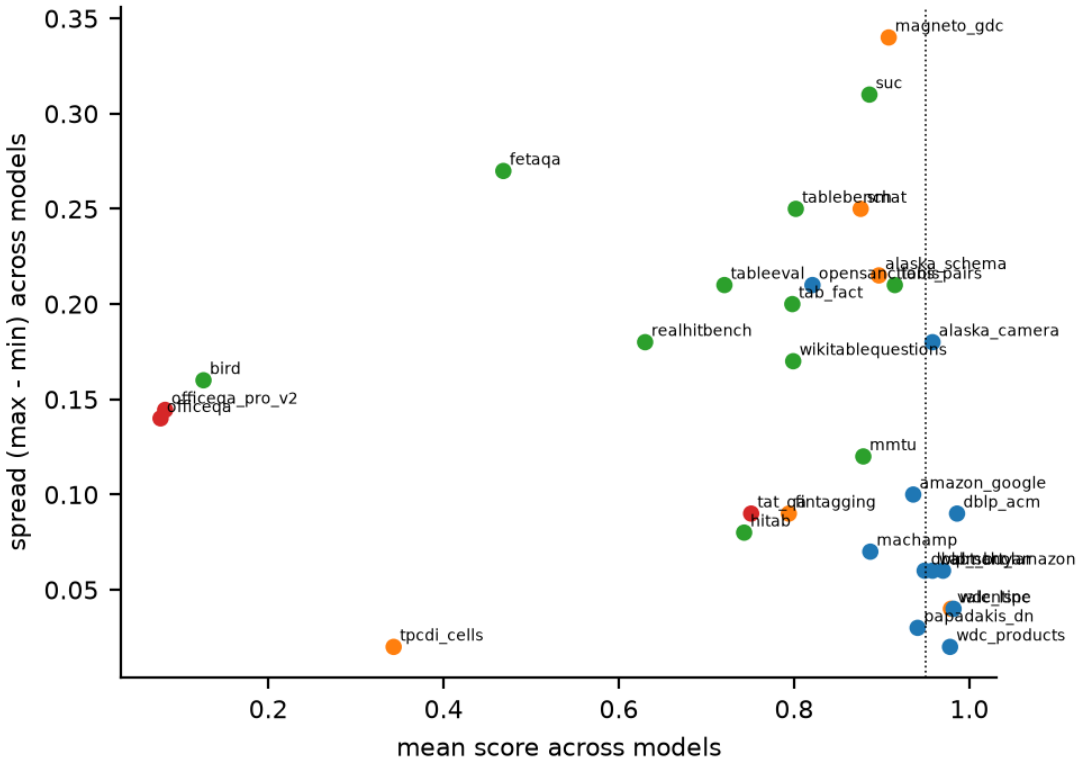


Figure 18: Dataset diagnostics: mean score against spread across models; dotted line marks the ceiling flag.

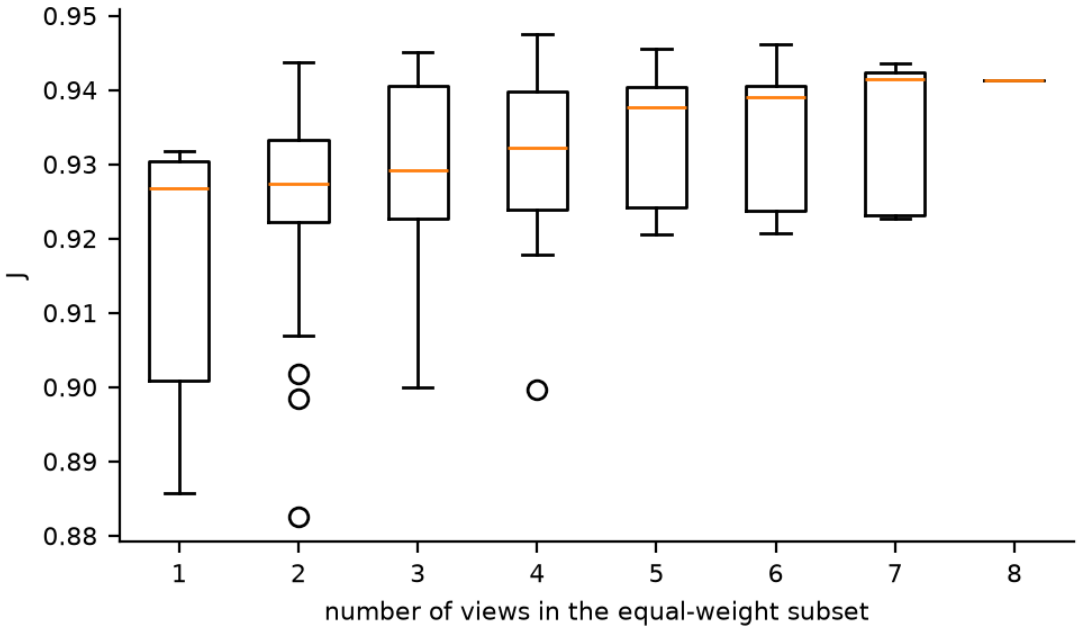


Figure 19: Objective J of the 255 equal-weight subsets by subset size.

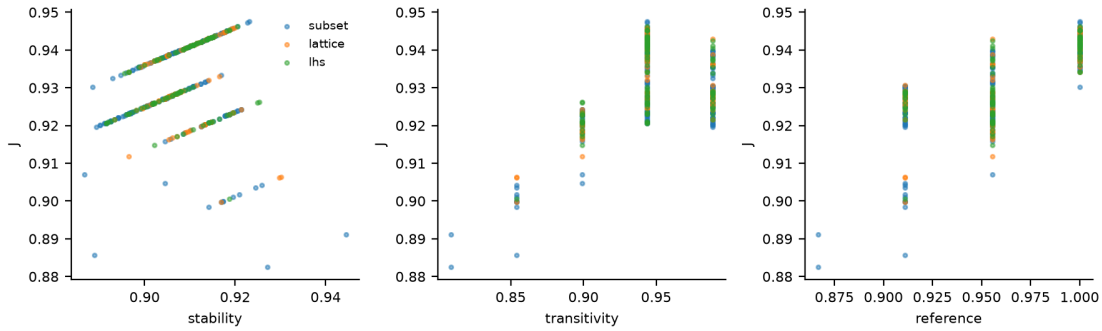


Figure 20: Objective J against each of its three terms (stability, transitivity, reference) for every candidate mix.

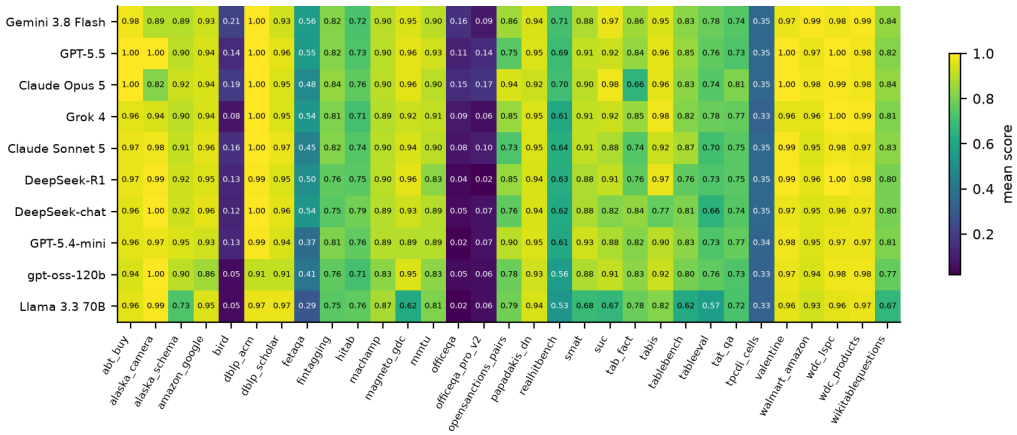


Figure 21: Mean score per model and dataset under the headline rule.

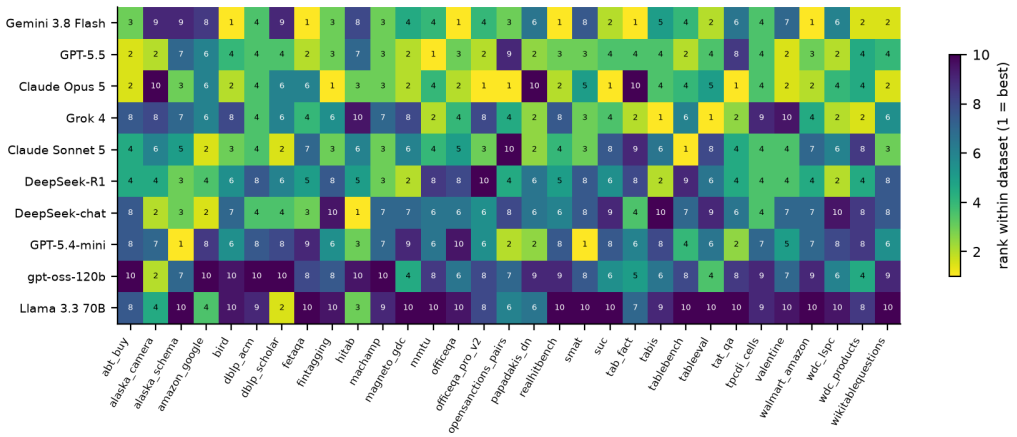


Figure 22: Rank per model and dataset under the headline rule.

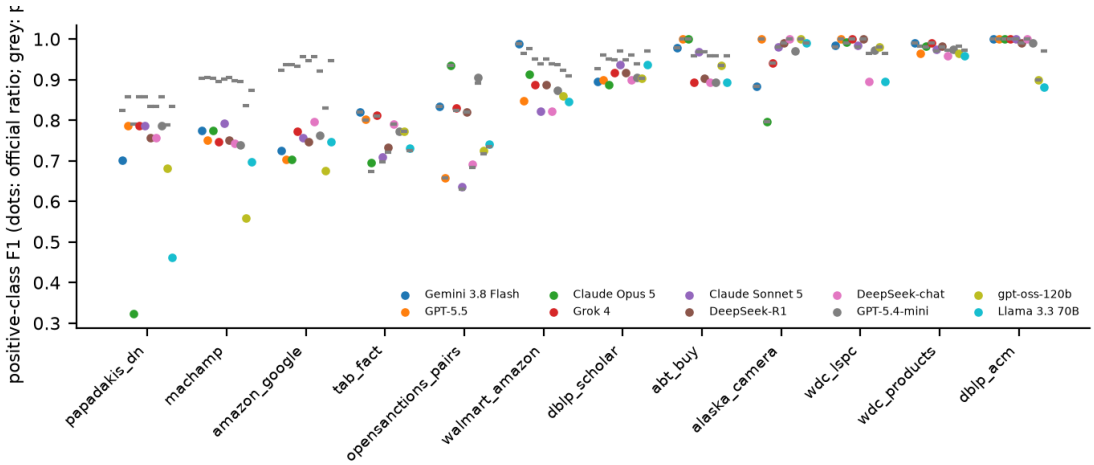


Figure 23: Positive-class F1 on the balanced pool (grey) and at the official match ratio (coloured) per model and matching dataset.

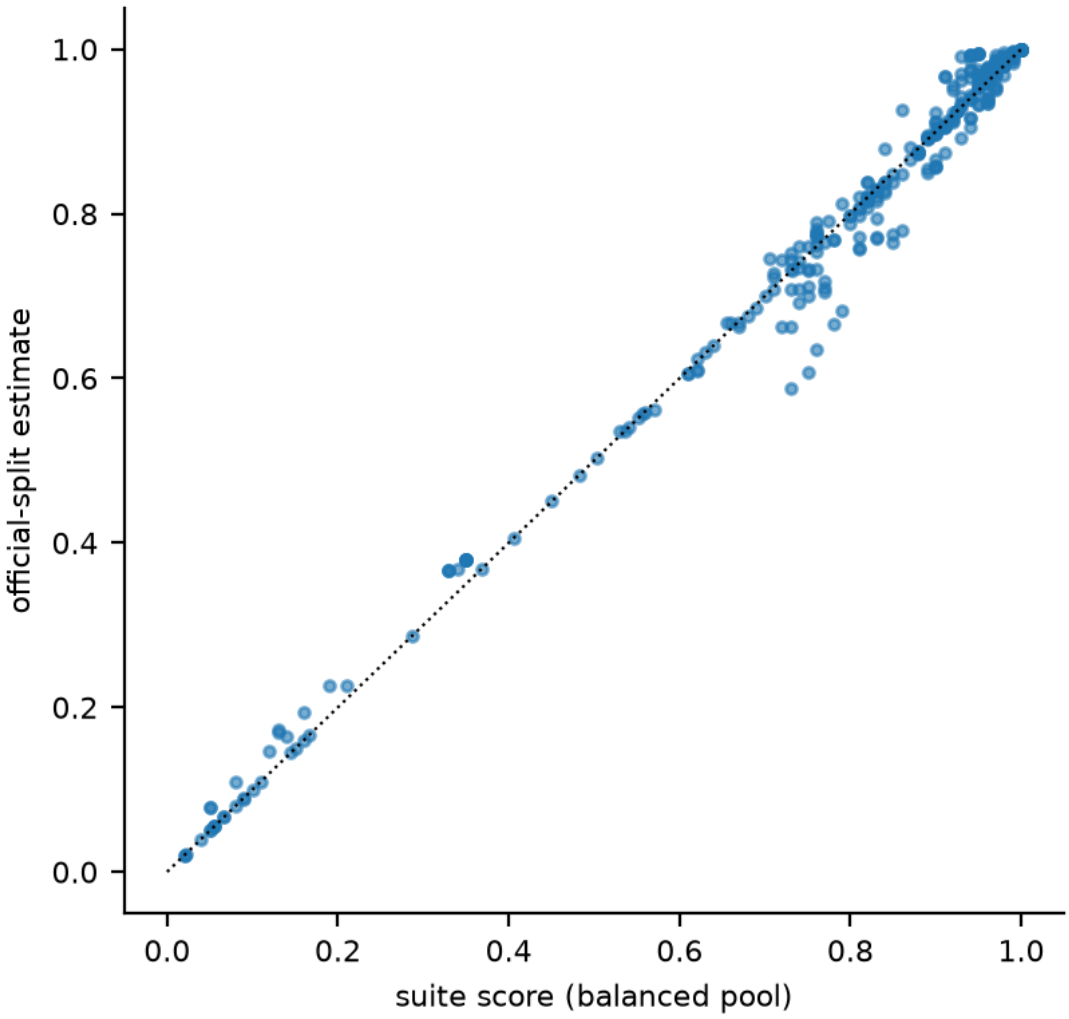


Figure 24: Suite score against post-stratified official-split estimate for every (dataset, model) pair under the headline rule.

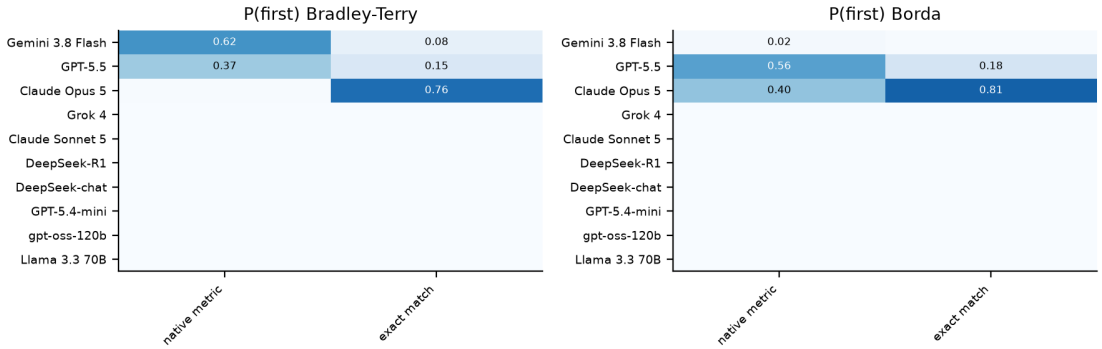


Figure 25: Bootstrap probability of first place under every rule, by pooling method.

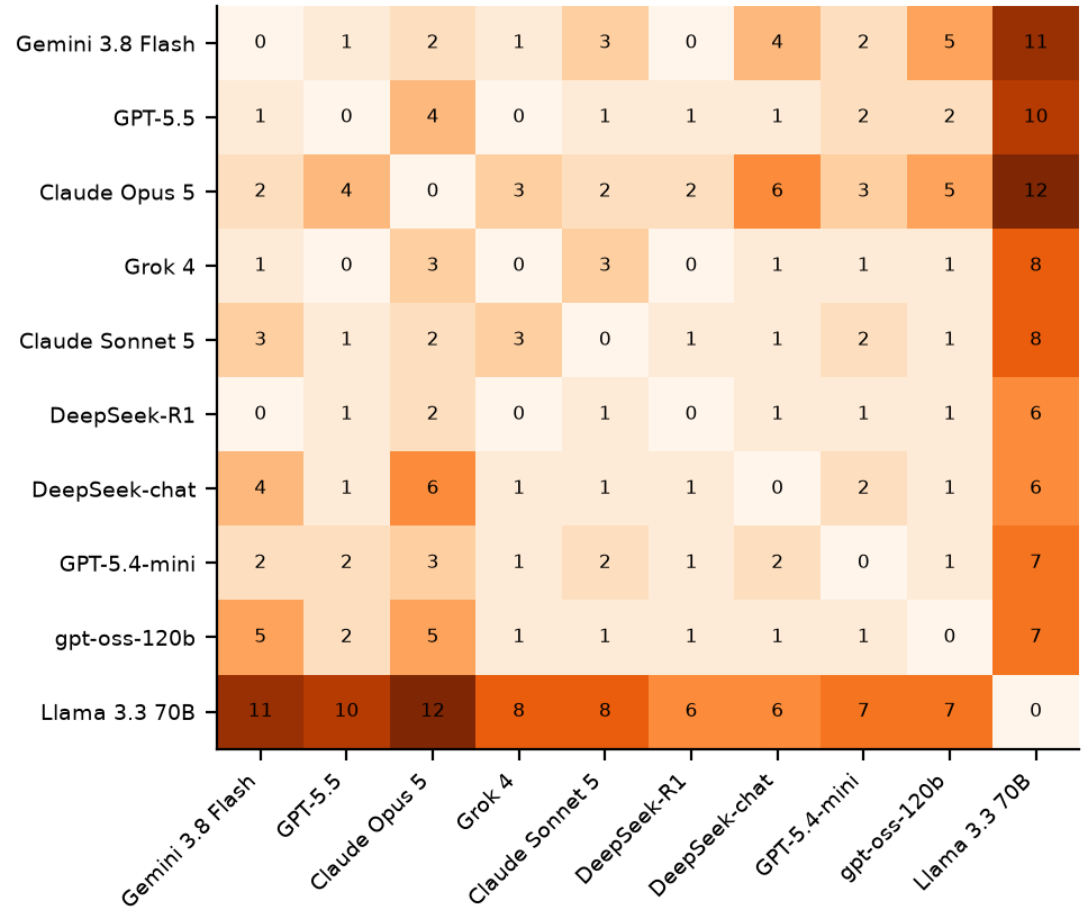


Figure 26: Number of datasets (of 31) in which the per-item paired test separates each pair of models under the headline rule (Holm within dataset).

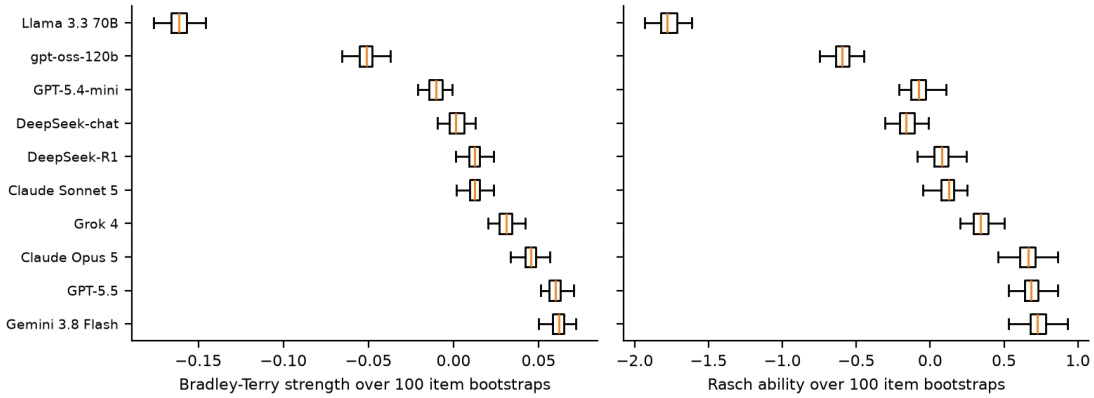


Figure 27: Bootstrap distributions of the Bradley–Terry strengths and Rasch abilities (100 item resamples).

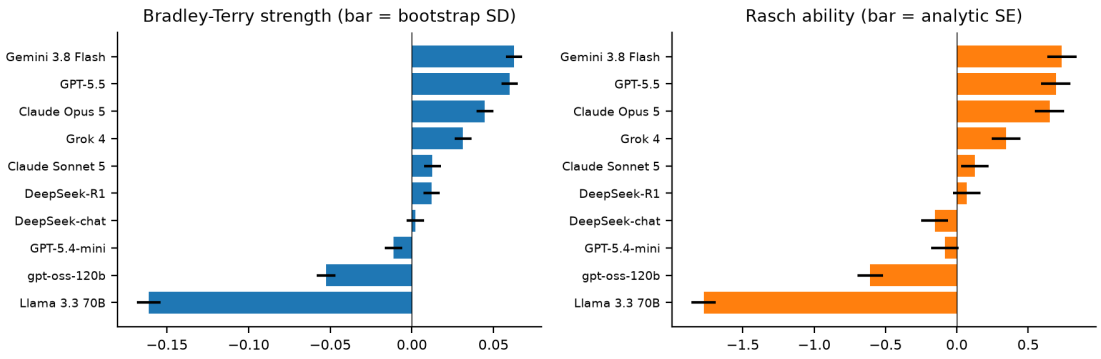


Figure 28: Fitted Bradley–Terry strengths and Rasch abilities with their uncertainty.

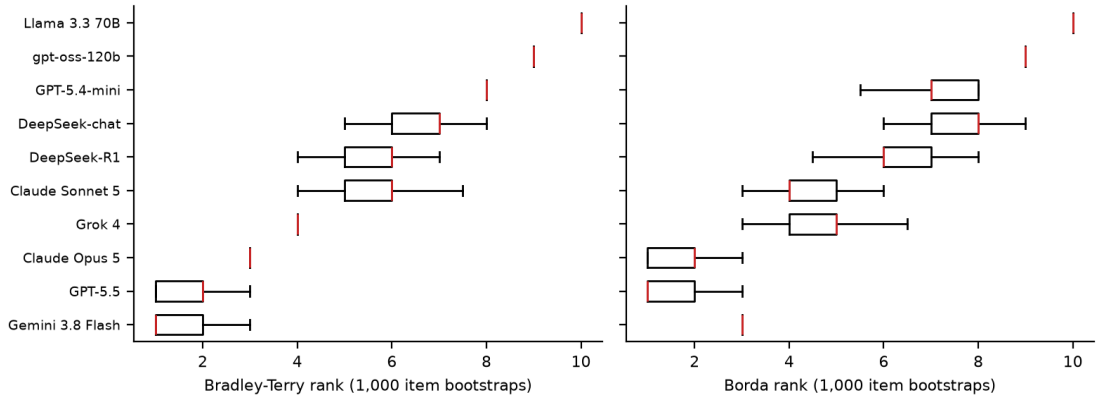


Figure 29: Bootstrap distribution of each model's rank under the headline rule (native metric).

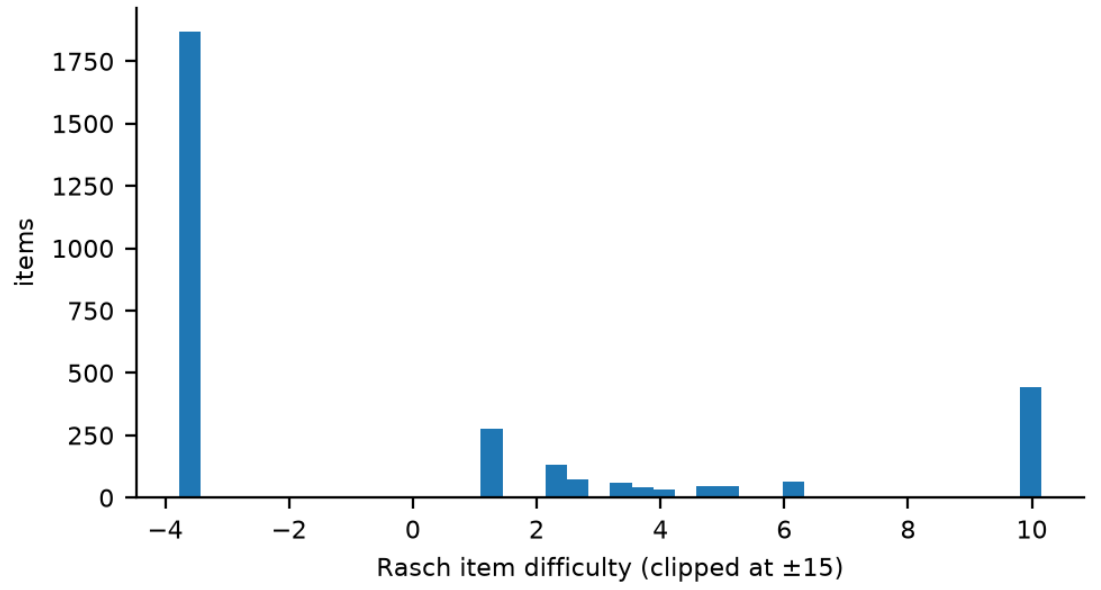


Figure 30: Distribution of Rasch item difficulties over the 3,083 items.

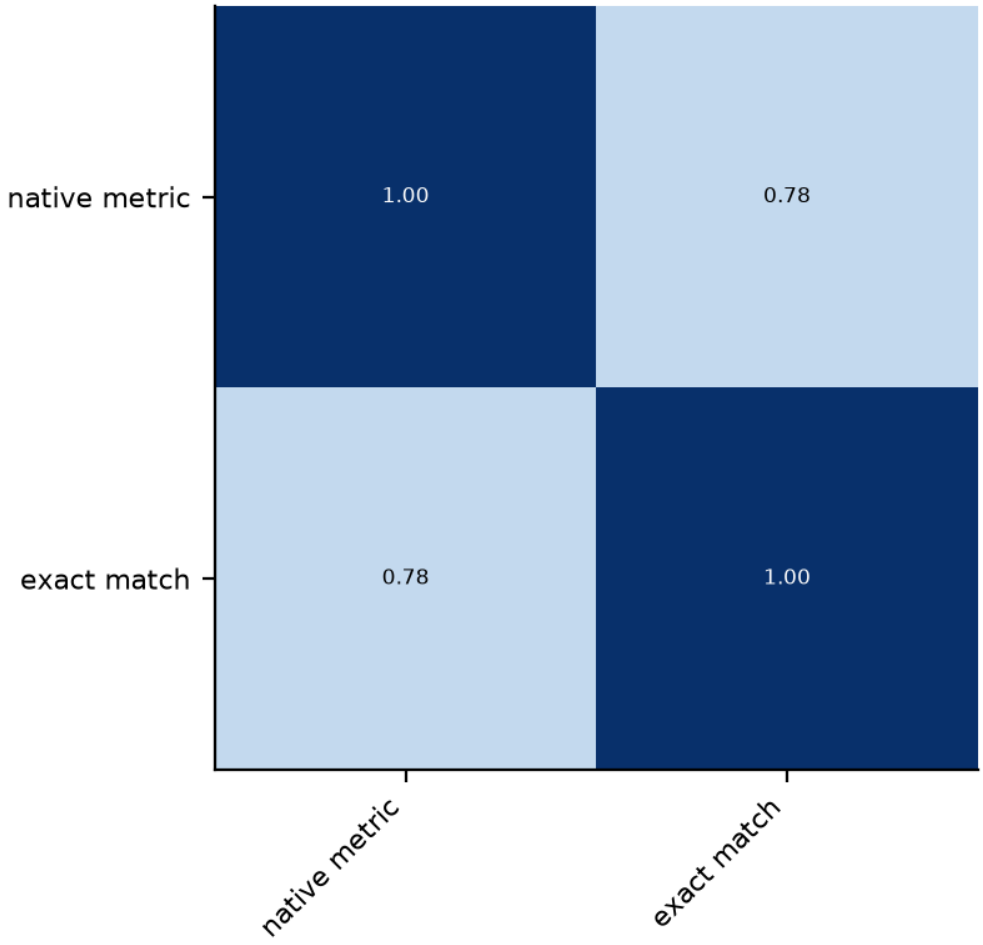


Figure 31: Kendall τ between the Bradley–Terry rankings of each pair of rules.

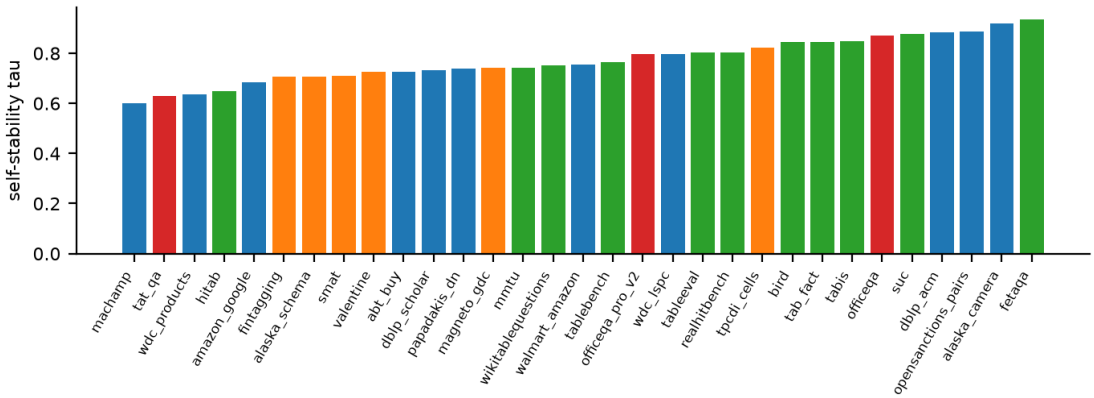


Figure 32: Self-stability of each dataset's model ranking (Kendall τ between bootstrap halves).

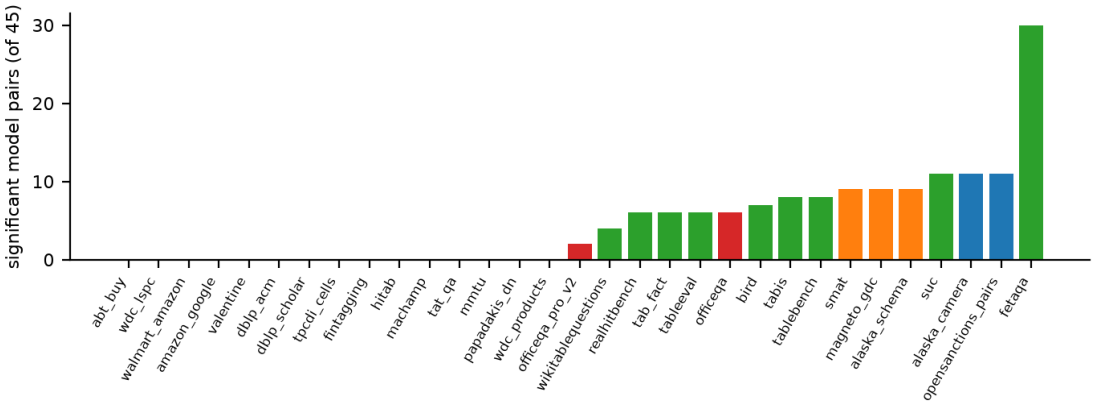


Figure 33: How many of the 45 model pairs each dataset separates at the item level under the headline rule (colour = family).

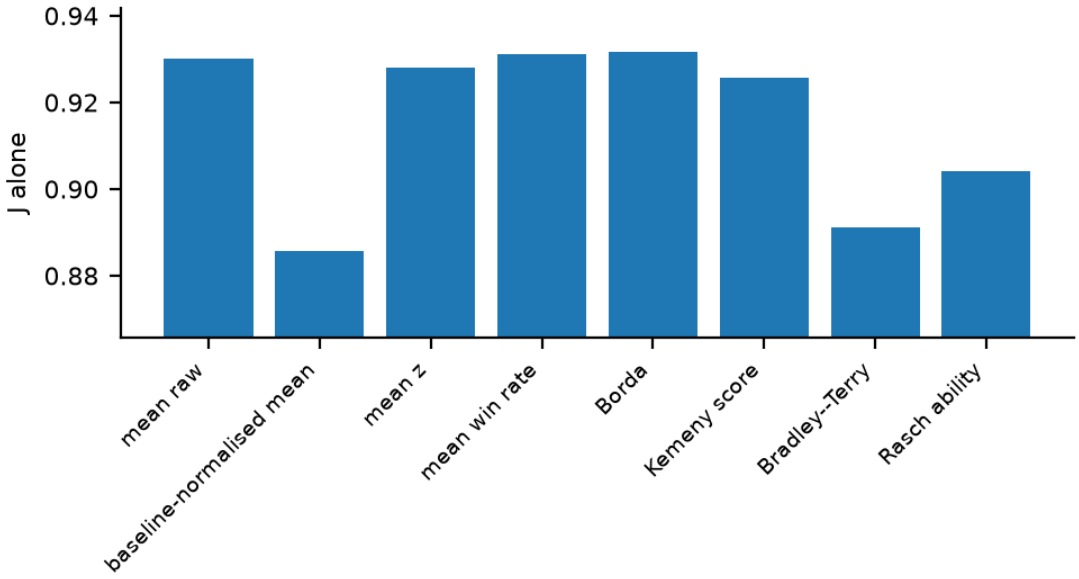


Figure 34: Objective J of each view used alone.

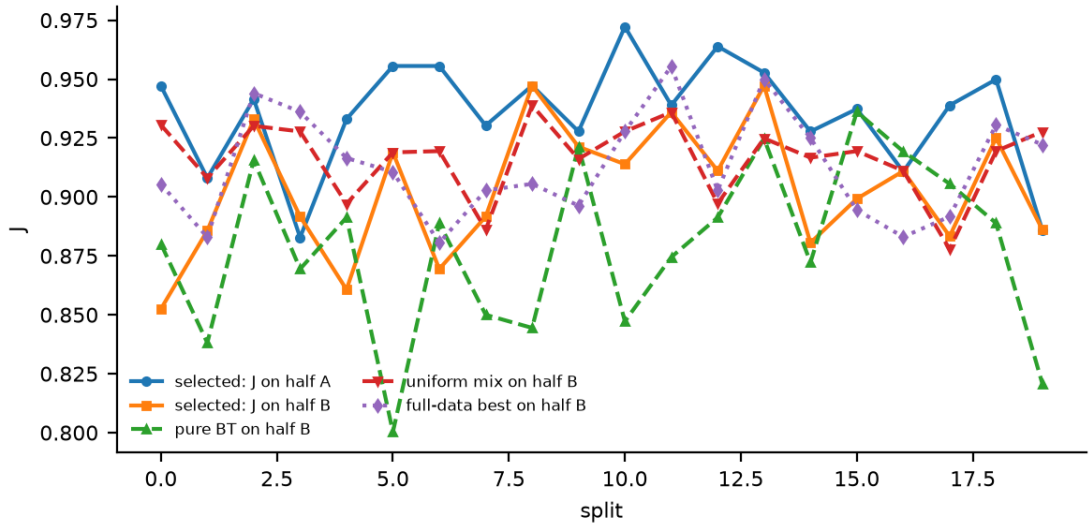


Figure 35: Split-half honesty check: the mix selected on half A scored on both halves, against pure Bradley–Terry, the uniform mix and the full-data best on half B.

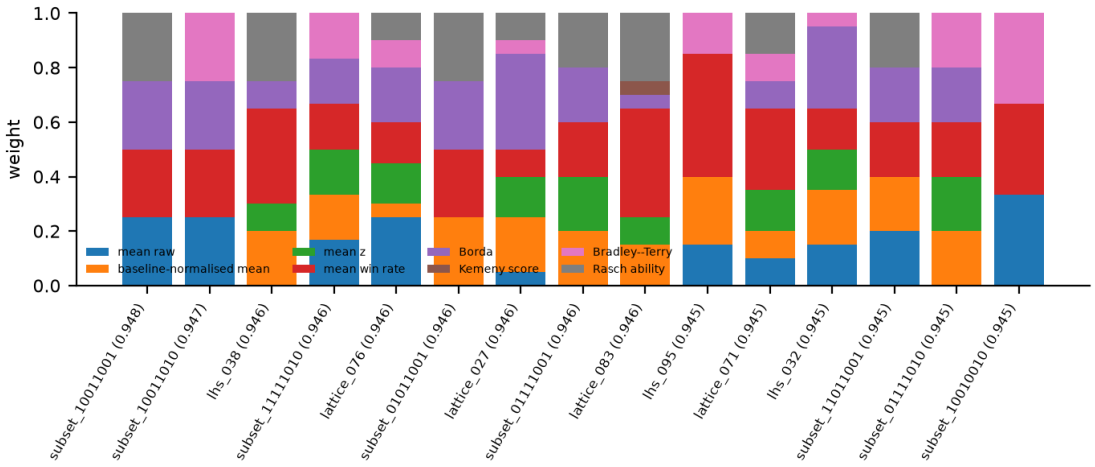


Figure 36: View weights of the fifteen best candidate mixes (label and J on the axis).

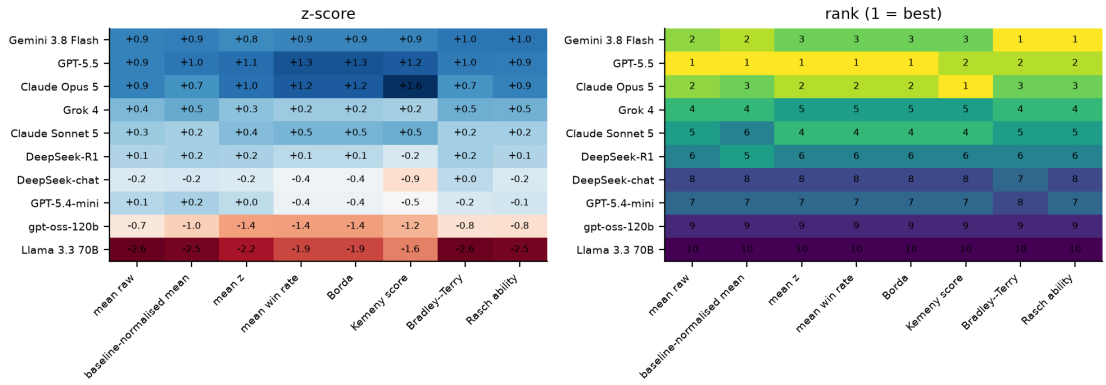


Figure 37: Each view's z-score (left) and rank (right) per model on native-metric scores.

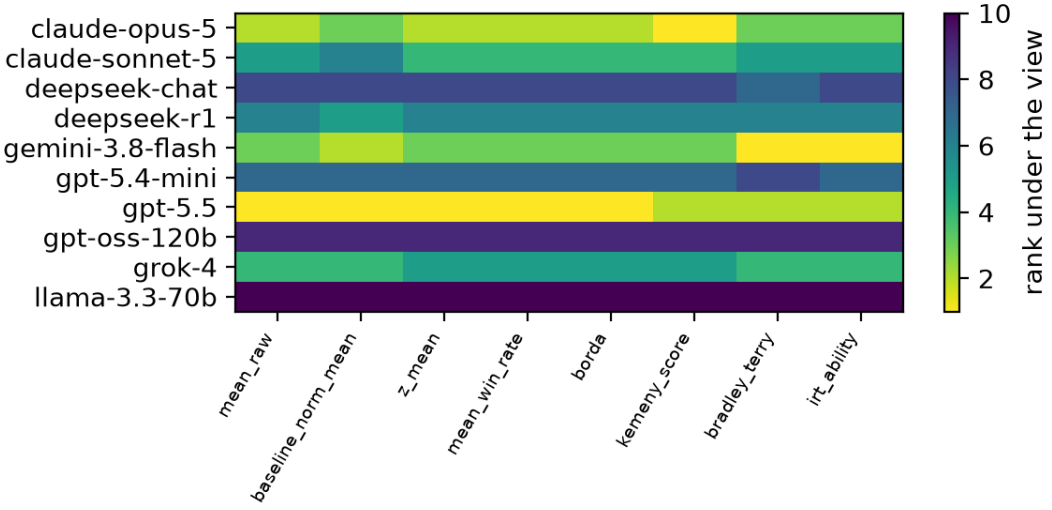


Figure 38: The eight single views: z-scored value per model.

B All tables

Every saved table, one file each under `tables/`; include any in the main text with `\input`.

Table 8: Best equal-weight subset for each subset size.

n_views	label	J	stability	transitivity	reference
1	subset_00001000	0.932	0.914	0.944	0.956
2	subset_00010001	0.944	0.916	0.944	1.000
3	subset_10010010	0.945	0.918	0.944	1.000
4	subset_10011001	0.948	0.923	0.944	1.000
5	subset_01111001	0.946	0.919	0.944	1.000
6	subset_11111010	0.946	0.920	0.944	1.000
7	subset_11111011	0.944	0.915	0.944	1.000
8	subset_11111111	0.941	0.911	0.944	1.000

Table 9: Sampled designs of 100 mixes: coverage diagnostics.

design	n	K	min_ pairwise_ L1	mean_ nearest_ L1	centred_L2_ discrepancy	mean_distinct_values_per_ weight	share_with_a_zero_ weight
lattice	100	8	0.200	0.401	1.993	11.625	0.930
lhs	100	8	0.100	0.366	2.071	10.500	0.760

Table 10: Dataset diagnostics under the headline rule.

dataset	family	mean_score	spread_ across_models	floor	ceiling	share_ constant_ bootstrap_ rankings	ranking_self_ stability_tau	tau_to_ others_mean	n_items	exclude_ from_ objective
abt_buy	EM	0.970	0.060	no	no	0.000	0.726	0.369	100	no
alaska_camera	EM	0.958	0.180	no	no	0.000	0.919	-0.219	100	no
alaska_schema	SM	0.897	0.215	no	no	0.000	0.707	0.055	93	no
amazon_google	EM	0.936	0.100	no	no	0.000	0.684	-0.018	100	no
bird	TQA	0.126	0.160	no	no	0.000	0.844	0.364	100	no
dblp_acm	EM	0.986	0.090	no	no	0.000	0.884	0.358	100	no
dblp_scholar	EM	0.949	0.060	no	no	0.000	0.733	-0.025	100	no
fetaqa	TQA	0.468	0.270	no	no	0.000	0.937	0.282	100	no
fintagging	SM	0.794	0.090	no	no	0.000	0.707	0.376	100	no
hitab	TQA	0.743	0.080	no	no	0.000	0.647	-0.091	100	no
machamp	EM	0.887	0.070	no	no	0.000	0.600	0.396	100	no
magneto_gdc	SM	0.908	0.340	no	no	0.000	0.741	0.265	100	no
mmtu	TQA	0.879	0.120	no	no	0.000	0.743	0.362	100	no
officeqa	FIN	0.077	0.140	no	no	0.000	0.870	0.325	100	no
officeqa_pro_v2	FIN	0.082	0.144	no	no	0.000	0.795	0.303	90	no
opensanctions_pairs	EM	0.821	0.210	no	no	0.000	0.888	0.025	100	no
papadakis_dn	EM	0.941	0.030	no	no	0.002	0.739	0.074	100	no
realhitbench	TQA	0.630	0.180	no	no	0.000	0.803	0.385	100	no
smat	SM	0.876	0.250	no	no	0.000	0.711	0.196	100	no
suc	TQA	0.886	0.310	no	no	0.000	0.879	0.322	100	no
tab_fact	TQA	0.798	0.200	no	no	0.000	0.845	-0.002	100	no
tabis	TQA	0.915	0.210	no	no	0.000	0.849	0.239	100	no
tablebench	TQA	0.802	0.250	no	no	0.000	0.766	0.316	100	no
tableeval	TQA	0.721	0.210	no	no	0.000	0.803	0.192	100	no
tat_qa	FIN	0.751	0.090	no	no	0.000	0.628	0.195	100	no
tpcdi_cells	SM	0.343	0.020	no	no	0.058	0.821	0.329	100	no
valentine	SM	0.979	0.040	no	yes	0.000	0.725	0.296	100	yes
walmart_amazon	EM	0.958	0.060	no	no	0.000	0.755	0.386	100	no
wdc_lspc	EM	0.982	0.040	no	yes	0.004	0.796	0.268	100	yes
wdc_products	EM	0.978	0.020	no	yes	0.006	0.637	0.188	100	yes
wikitablequestions	TQA	0.799	0.170	no	no	0.000	0.751	0.393	100	no

Table 11: Full leaderboard under rule exact match; weights: –.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Claude Opus 5	0.047	0.000	1.000	1.000	3.000	0.764	1.000	1.000	2.000	0.812	1.000	1.000	1.000	1.000	1.000	0.768
GPT-5.5	0.039	0.016	2.000	1.000	3.000	0.152	2.000	1.000	3.000	0.177	2.000	2.000	2.000	2.000	2.000	0.764
Gemini 3.8 Flash	0.037	0.016	3.000	1.000	3.000	0.084	3.000	2.000	5.000	0.004	3.000	3.000	3.000	3.000	3.000	0.763
Grok 4	0.020	0.016	4.000	4.000	6.000	0.000	5.000	3.000	6.012	0.000	5.000	5.000	5.000	5.000	5.000	0.754
Claude Sonnet 5	0.013	0.016	5.000	4.000	7.000	0.000	4.000	3.000	6.000	0.000	4.000	4.000	4.000	4.000	4.000	0.751
GPT-5.4-mini	0.010	0.016	6.000	4.000	7.000	0.000	8.000	6.000	8.000	0.000	7.000	7.000	7.000	8.000	8.000	0.749
DeepSeek-R1	0.008	0.016	7.000	5.000	7.000	0.000	6.000	4.000	8.000	0.000	6.000	6.000	6.000	6.000	6.000	0.748
DeepSeek-chat	-0.014	0.016	8.000	8.000	9.000	0.000	7.000	6.000	8.000	0.000	8.000	8.000	8.000	7.000	7.000	0.737
gpt-oss-120b	-0.033	0.016	9.000	8.000	9.000	0.000	9.000	9.000	9.000	0.000	9.000	9.000	9.000	9.000	9.000	0.728
Llama 3.3 70B	-0.128	0.016	10.000	10.000	10.000	0.000	10.000	10.000	10.000	0.000	10.000	10.000	10.000	10.000	10.000	0.680

Table 12: Leaderboard under the headline rule (native metric).

rank	model	BT strength	SE	Borda rank	Borda 95%	P(first) BT	mean score
1	Gemini 3.8 Flash	0.063	0.016	3	[2, 4]	0.620	0.792
2	GPT-5.5	0.060	0.016	1	[1, 3]	0.370	0.793
3	Claude Opus 5	0.045	0.000	2	[1, 3]	0.000	0.792
4	Grok 4	0.031	0.016	5	[3, 6]	0.000	0.780
5	Claude Sonnet 5	0.012	0.016	4	[3, 6]	0.000	0.776
6	DeepSeek-R1	0.012	0.016	6	[4, 8]	0.000	0.772
7	DeepSeek-chat	0.002	0.016	8	[6, 8]	0.000	0.763
8	GPT-5.4-mini	-0.011	0.016	7	[6, 8]	0.000	0.771
9	gpt-oss-120b	-0.053	0.016	9	[9, 9]	0.000	0.750
10	Llama 3.3 70B	-0.161	0.016	10	[10, 10]	0.000	0.701

Table 13: Full leaderboard under rule native metric; weights: –.

model	bt__strength	bt__se__vs__ref	rank__bt	bt__rank__lo95	bt__rank__hi95	p__first__bt	rank__borda	borda__rank__lo95	borda__rank__hi95	p__first__borda	rank__kemeny	rank__copeland	rank__mean__z	rank__rrf	rank__family__balanced__z	mean__score
Gemini 3.8 Flash	0.063	0.016	1.000	1.000	2.000	0.620	3.000	2.000	4.000	0.021	3.000	3.000	3.000	3.000	3.000	0.792
GPT-5.5	0.060	0.016	2.000	1.000	3.000	0.374	1.000	1.000	3.000	0.560	2.000	2.000	1.000	1.000	2.000	0.793
Claude Opus 5	0.045	0.000	3.000	2.000	4.000	0.005	2.000	1.000	3.000	0.404	1.000	1.000	2.000	2.000	1.000	0.792
Grok 4	0.031	0.016	4.000	3.000	5.000	0.000	5.000	3.000	6.000	0.000	5.000	4.500	5.000	5.000	5.000	0.780
Claude Sonnet 5	0.012	0.016	5.000	4.000	7.000	0.000	4.000	3.000	6.000	0.000	4.000	4.500	4.000	4.000	4.000	0.776
DeepSeek-R1	0.012	0.016	6.000	5.000	7.000	0.000	6.000	4.000	8.000	0.000	6.000	6.000	6.000	6.000	7.000	0.772
DeepSeek-chat	0.002	0.016	7.000	5.000	8.000	0.000	8.000	6.000	8.000	0.000	8.000	8.000	8.000	8.000	8.000	0.763
GPT-5.4-mini	-0.011	0.016	8.000	7.000	8.000	0.000	7.000	6.000	8.000	0.000	7.000	7.000	7.000	7.000	6.000	0.771
gpt-oss-120b	-0.053	0.016	9.000	9.000	9.000	0.000	9.000	9.000	9.000	0.000	9.000	9.000	9.000	9.000	9.000	0.750
Llama 3.3 70B	-0.161	0.016	10.000	10.000	10.000	0.000	10.000	10.000	10.000	0.000	10.000	10.000	10.000	10.000	10.000	0.701

Table 14: Official-split estimates under the headline rule for every dataset and model.

dataset	model	suite_mean	official_ estimate	official_lo95	official_hi95	effective_n	suite_f1_pos	official_f1_ pos	official_f1_ lo95	official_f1_ hi95	official_ positive_ share
abt_buy	Claude Opus 5	1.000	1.000	1.000	1.000	63.075	1.000	1.000	1.000	1.000	0.108
abt_buy	Claude Sonnet 5	0.970	0.993	0.985	1.000	63.075	0.968	0.968	0.925	1.000	0.108
abt_buy	DeepSeek-chat	0.960	0.976	0.933	0.998	63.075	0.958	0.893	0.735	0.990	0.108
abt_buy	DeepSeek-R1	0.970	0.978	0.931	1.000	63.075	0.969	0.904	0.755	1.000	0.108
abt_buy	Gemini 3.8 Flash	0.980	0.996	0.988	1.000	63.075	0.979	0.979	0.944	1.000	0.108
abt_buy	GPT-5.4-mini	0.960	0.976	0.937	0.998	63.075	0.958	0.893	0.750	0.990	0.108
abt_buy	GPT-5.5	1.000	1.000	1.000	1.000	63.075	1.000	1.000	1.000	1.000	0.108
abt_buy	gpt-oss-120b	0.940	0.987	0.974	0.996	63.075	0.935	0.935	0.879	0.981	0.108
abt_buy	Grok 4	0.960	0.976	0.934	0.998	63.075	0.958	0.893	0.745	0.990	0.108
abt_buy	Llama 3.3 70B	0.960	0.976	0.937	0.998	63.075	0.958	0.893	0.735	0.992	0.108
alaska_camera	Claude Opus 5	0.820	0.820	0.740	0.890	100.000	0.795	0.795	0.687	0.881	0.500
alaska_camera	Claude Sonnet 5	0.980	0.980	0.950	1.000	100.000	0.981	0.981	0.951	1.000	0.500
alaska_camera	DeepSeek-chat	1.000	1.000	1.000	1.000	100.000	1.000	1.000	1.000	1.000	0.500
alaska_camera	DeepSeek-R1	0.990	0.990	0.970	1.000	100.000	0.990	0.990	0.969	1.000	0.500
alaska_camera	Gemini 3.8 Flash	0.890	0.890	0.830	0.950	100.000	0.884	0.884	0.805	0.943	0.500
alaska_camera	GPT-5.4-mini	0.970	0.970	0.930	1.000	100.000	0.971	0.971	0.929	1.000	0.500
alaska_camera	GPT-5.5	1.000	1.000	1.000	1.000	100.000	1.000	1.000	1.000	1.000	0.500
alaska_camera	gpt-oss-120b	1.000	1.000	1.000	1.000	100.000	1.000	1.000	1.000	1.000	0.500
alaska_camera	Grok 4	0.940	0.940	0.890	0.980	100.000	0.940	0.940	0.886	0.982	0.500
alaska_camera	Llama 3.3 70B	0.990	0.990	0.970	1.000	100.000	0.990	0.990	0.967	1.000	0.500
alaska_schema	Claude Opus 5	0.925	0.925	0.871	0.968	93.000	–	–	–	–	–
alaska_schema	Claude Sonnet 5	0.914	0.914	0.860	0.968	93.000	–	–	–	–	–
alaska_schema	DeepSeek-chat	0.925	0.925	0.860	0.978	93.000	–	–	–	–	–
alaska_schema	DeepSeek-R1	0.925	0.925	0.871	0.978	93.000	–	–	–	–	–
alaska_schema	Gemini 3.8 Flash	0.892	0.892	0.828	0.946	93.000	–	–	–	–	–
alaska_schema	GPT-5.4-mini	0.946	0.946	0.892	0.979	93.000	–	–	–	–	–
alaska_schema	GPT-5.5	0.903	0.903	0.839	0.957	93.000	–	–	–	–	–
alaska_schema	gpt-oss-120b	0.903	0.903	0.839	0.968	93.000	–	–	–	–	–
alaska_schema	Grok 4	0.903	0.903	0.839	0.957	93.000	–	–	–	–	–
alaska_schema	Llama 3.3 70B	0.731	0.731	0.645	0.817	93.000	–	–	–	–	–
amazon_google	Claude Opus 5	0.940	0.916	0.841	0.981	67.152	0.936	0.704	0.526	0.907	0.102
amazon_google	Claude Sonnet 5	0.960	0.935	0.865	0.985	67.152	0.957	0.758	0.587	0.933	0.102
amazon_google	DeepSeek-chat	0.960	0.949	0.886	0.998	67.152	0.957	0.796	0.628	0.992	0.102
amazon_google	DeepSeek-R1	0.950	0.932	0.867	0.985	67.152	0.946	0.747	0.583	0.930	0.102
amazon_google	Gemini 3.8 Flash	0.930	0.928	0.863	0.979	67.152	0.923	0.725	0.561	0.926	0.102
amazon_google	GPT-5.4-mini	0.930	0.942	0.876	0.990	67.152	0.921	0.762	0.590	0.962	0.102

amazon_google	GPT-5.5	0.940	0.916	0.840	0.981	67.152	0.936	0.704	0.529	0.917	0.102
amazon_google	gpt-oss-120b	0.860	0.926	0.865	0.975	67.152	0.829	0.676	0.502	0.855	0.102
amazon_google	Grok 4	0.940	0.944	0.886	0.994	67.152	0.933	0.773	0.606	0.968	0.102
amazon_google	Llama 3.3 70B	0.950	0.932	0.867	0.985	67.152	0.946	0.747	0.583	0.932	0.102
bird	Claude Opus 5	0.190	0.226	0.125	0.324	74.082	–	–	–	–	–
bird	Claude Sonnet 5	0.160	0.194	0.108	0.299	74.082	–	–	–	–	–
bird	DeepSeek-chat	0.120	0.147	0.066	0.230	74.082	–	–	–	–	–
bird	DeepSeek-R1	0.130	0.170	0.082	0.261	74.082	–	–	–	–	–
bird	Gemini 3.8 Flash	0.210	0.226	0.132	0.326	74.082	–	–	–	–	–
bird	GPT-5.4-mini	0.130	0.173	0.091	0.260	74.082	–	–	–	–	–
bird	GPT-5.5	0.140	0.164	0.086	0.256	74.082	–	–	–	–	–
bird	gpt-oss-120b	0.050	0.079	0.018	0.150	74.082	–	–	–	–	–
bird	Grok 4	0.080	0.109	0.042	0.188	74.082	–	–	–	–	–
bird	Llama 3.3 70B	0.050	0.079	0.018	0.157	74.082	–	–	–	–	–
dblp_acm	Claude Opus 5	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	Claude Sonnet 5	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	DeepSeek-chat	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	DeepSeek-R1	0.990	0.996	0.988	1.000	72.166	0.990	0.990	0.966	1.000	0.180
dblp_acm	Gemini 3.8 Flash	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	GPT-5.4-mini	0.990	0.996	0.987	1.000	72.166	0.990	0.990	0.964	1.000	0.180
dblp_acm	GPT-5.5	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	gpt-oss-120b	0.910	0.967	0.943	0.987	72.166	0.899	0.899	0.819	0.958	0.180
dblp_acm	Grok 4	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	Llama 3.3 70B	0.970	0.952	0.893	1.000	72.166	0.970	0.882	0.752	1.000	0.180
dblp_scholar	Claude Opus 5	0.950	0.956	0.906	0.993	70.463	0.950	0.888	0.777	0.983	0.186
dblp_scholar	Claude Sonnet 5	0.970	0.976	0.941	1.000	70.463	0.970	0.937	0.836	1.000	0.186
dblp_scholar	DeepSeek-chat	0.960	0.959	0.911	0.996	70.463	0.961	0.898	0.793	0.991	0.186
dblp_scholar	DeepSeek-R1	0.950	0.969	0.930	0.993	70.463	0.949	0.917	0.811	0.990	0.186
dblp_scholar	Gemini 3.8 Flash	0.930	0.961	0.920	0.990	70.463	0.928	0.895	0.795	0.972	0.186
dblp_scholar	GPT-5.4-mini	0.940	0.965	0.922	0.992	70.463	0.939	0.906	0.788	0.980	0.186
dblp_scholar	GPT-5.5	0.960	0.959	0.907	0.996	70.463	0.961	0.898	0.778	0.991	0.186
dblp_scholar	gpt-oss-120b	0.910	0.967	0.943	0.987	70.463	0.903	0.903	0.838	0.962	0.186
dblp_scholar	Grok 4	0.950	0.969	0.930	0.996	70.463	0.949	0.917	0.822	0.990	0.186
dblp_scholar	Llama 3.3 70B	0.970	0.976	0.938	1.000	70.463	0.970	0.937	0.851	1.000	0.186
fetaqa	Claude Opus 5	0.483	0.483	0.443	0.522	100.000	–	–	–	–	–
fetaqa	Claude Sonnet 5	0.451	0.451	0.409	0.490	100.000	–	–	–	–	–
fetaqa	DeepSeek-chat	0.540	0.540	0.499	0.582	100.000	–	–	–	–	–
fetaqa	DeepSeek-R1	0.504	0.504	0.458	0.551	100.000	–	–	–	–	–
fetaqa	Gemini 3.8 Flash	0.557	0.557	0.517	0.597	100.000	–	–	–	–	–
fetaqa	GPT-5.4-mini	0.368	0.368	0.333	0.407	100.000	–	–	–	–	–
fetaqa	GPT-5.5	0.552	0.552	0.511	0.593	100.000	–	–	–	–	–
fetaqa	gpt-oss-120b	0.406	0.406	0.380	0.435	100.000	–	–	–	–	–

fetaqa	Grok 4	0.536	0.536	0.497	0.578	100.000	–	–	–	–	–
fetaqa	Llama 3.3 70B	0.287	0.287	0.249	0.324	100.000	–	–	–	–	–
fintagging	Claude Opus 5	0.840	0.880	0.787	0.958	51.769	–	–	–	–	–
fintagging	Claude Sonnet 5	0.820	0.818	0.714	0.916	51.769	–	–	–	–	–
fintagging	DeepSeek-chat	0.750	0.731	0.611	0.854	51.769	–	–	–	–	–
fintagging	DeepSeek-R1	0.760	0.790	0.676	0.889	51.769	–	–	–	–	–
fintagging	Gemini 3.8 Flash	0.820	0.838	0.732	0.926	51.769	–	–	–	–	–
fintagging	GPT-5.4-mini	0.810	0.759	0.633	0.874	51.769	–	–	–	–	–
fintagging	GPT-5.5	0.820	0.838	0.727	0.930	51.769	–	–	–	–	–
fintagging	gpt-oss-120b	0.760	0.732	0.613	0.847	51.769	–	–	–	–	–
fintagging	Grok 4	0.810	0.798	0.681	0.896	51.769	–	–	–	–	–
fintagging	Llama 3.3 70B	0.750	0.711	0.594	0.827	51.769	–	–	–	–	–
hitab	Claude Opus 5	0.760	0.782	0.702	0.858	94.537	–	–	–	–	–
hitab	Claude Sonnet 5	0.740	0.743	0.652	0.830	94.537	–	–	–	–	–
hitab	DeepSeek-chat	0.790	0.812	0.737	0.882	94.537	–	–	–	–	–
hitab	DeepSeek-R1	0.750	0.760	0.674	0.842	94.537	–	–	–	–	–
hitab	Gemini 3.8 Flash	0.720	0.745	0.662	0.827	94.537	–	–	–	–	–
hitab	GPT-5.4-mini	0.760	0.772	0.689	0.858	94.537	–	–	–	–	–
hitab	GPT-5.5	0.730	0.751	0.659	0.833	94.537	–	–	–	–	–
hitab	gpt-oss-120b	0.710	0.728	0.638	0.819	94.537	–	–	–	–	–
hitab	Grok 4	0.710	0.723	0.626	0.810	94.537	–	–	–	–	–
hitab	Llama 3.3 70B	0.760	0.777	0.690	0.855	94.537	–	–	–	–	–
machamp	Claude Opus 5	0.900	0.912	0.833	0.970	32.491	0.904	0.774	0.623	0.920	0.175
machamp	Claude Sonnet 5	0.900	0.923	0.844	0.971	32.491	0.902	0.793	0.645	0.922	0.175
machamp	DeepSeek-chat	0.890	0.892	0.801	0.959	32.491	0.897	0.742	0.572	0.897	0.175
machamp	DeepSeek-R1	0.900	0.897	0.807	0.959	32.491	0.906	0.751	0.593	0.897	0.175
machamp	Gemini 3.8 Flash	0.900	0.912	0.827	0.967	32.491	0.904	0.774	0.614	0.914	0.175
machamp	GPT-5.4-mini	0.890	0.893	0.805	0.956	32.491	0.895	0.739	0.574	0.897	0.175
machamp	GPT-5.5	0.900	0.897	0.805	0.963	32.491	0.906	0.751	0.593	0.904	0.175
machamp	gpt-oss-120b	0.830	0.794	0.642	0.902	32.491	0.835	0.558	0.372	0.774	0.175
machamp	Grok 4	0.890	0.896	0.812	0.960	32.491	0.895	0.747	0.594	0.887	0.175
machamp	Llama 3.3 70B	0.870	0.880	0.783	0.951	32.491	0.874	0.698	0.541	0.852	0.175
magneto_gdc	Claude Opus 5	0.960	0.962	0.921	0.992	93.027	–	–	–	–	–
magneto_gdc	Claude Sonnet 5	0.940	0.939	0.887	0.981	93.027	–	–	–	–	–
magneto_gdc	DeepSeek-chat	0.930	0.933	0.878	0.976	93.027	–	–	–	–	–
magneto_gdc	DeepSeek-R1	0.960	0.963	0.922	0.992	93.027	–	–	–	–	–
magneto_gdc	Gemini 3.8 Flash	0.950	0.954	0.914	0.991	93.027	–	–	–	–	–
magneto_gdc	GPT-5.4-mini	0.890	0.894	0.833	0.948	93.027	–	–	–	–	–
magneto_gdc	GPT-5.5	0.960	0.966	0.931	0.992	93.027	–	–	–	–	–
magneto_gdc	gpt-oss-120b	0.950	0.954	0.907	0.991	93.027	–	–	–	–	–
magneto_gdc	Grok 4	0.920	0.924	0.867	0.970	93.027	–	–	–	–	–
magneto_gdc	Llama 3.3 70B	0.620	0.624	0.519	0.714	93.027	–	–	–	–	–

mmtu	Claude Opus 5	0.900	0.858	0.766	0.936	86.538	–	–	–	–	–
mmtu	Claude Sonnet 5	0.900	0.858	0.771	0.941	86.538	–	–	–	–	–
mmtu	DeepSeek-chat	0.890	0.849	0.767	0.933	86.538	–	–	–	–	–
mmtu	DeepSeek-R1	0.830	0.772	0.668	0.872	86.538	–	–	–	–	–
mmtu	Gemini 3.8 Flash	0.900	0.856	0.762	0.937	86.538	–	–	–	–	–
mmtu	GPT-5.4-mini	0.890	0.855	0.768	0.931	86.538	–	–	–	–	–
mmtu	GPT-5.5	0.930	0.892	0.804	0.969	86.538	–	–	–	–	–
mmtu	gpt-oss-120b	0.830	0.771	0.671	0.861	86.538	–	–	–	–	–
mmtu	Grok 4	0.910	0.875	0.788	0.950	86.538	–	–	–	–	–
mmtu	Llama 3.3 70B	0.810	0.757	0.656	0.860	86.538	–	–	–	–	–
officeqa	Claude Opus 5	0.150	0.150	0.090	0.220	100.000	–	–	–	–	–
officeqa	Claude Sonnet 5	0.080	0.080	0.030	0.140	100.000	–	–	–	–	–
officeqa	DeepSeek-chat	0.050	0.050	0.010	0.100	100.000	–	–	–	–	–
officeqa	DeepSeek-R1	0.040	0.040	0.010	0.080	100.000	–	–	–	–	–
officeqa	Gemini 3.8 Flash	0.160	0.160	0.090	0.230	100.000	–	–	–	–	–
officeqa	GPT-5.4-mini	0.020	0.020	0.000	0.050	100.000	–	–	–	–	–
officeqa	GPT-5.5	0.110	0.110	0.050	0.170	100.000	–	–	–	–	–
officeqa	gpt-oss-120b	0.050	0.050	0.010	0.100	100.000	–	–	–	–	–
officeqa	Grok 4	0.090	0.090	0.040	0.150	100.000	–	–	–	–	–
officeqa	Llama 3.3 70B	0.020	0.020	0.000	0.050	100.000	–	–	–	–	–
officeqa_pro_v2	Claude Opus 5	0.167	0.167	0.089	0.244	90.000	–	–	–	–	–
officeqa_pro_v2	Claude Sonnet 5	0.100	0.100	0.044	0.167	90.000	–	–	–	–	–
officeqa_pro_v2	DeepSeek-chat	0.067	0.067	0.022	0.122	90.000	–	–	–	–	–
officeqa_pro_v2	DeepSeek-R1	0.022	0.022	0.000	0.056	90.000	–	–	–	–	–
officeqa_pro_v2	Gemini 3.8 Flash	0.089	0.089	0.033	0.144	90.000	–	–	–	–	–
officeqa_pro_v2	GPT-5.4-mini	0.067	0.067	0.022	0.122	90.000	–	–	–	–	–
officeqa_pro_v2	GPT-5.5	0.144	0.144	0.078	0.222	90.000	–	–	–	–	–
officeqa_pro_v2	gpt-oss-120b	0.056	0.056	0.011	0.100	90.000	–	–	–	–	–
officeqa_pro_v2	Grok 4	0.056	0.056	0.011	0.100	90.000	–	–	–	–	–
officeqa_pro_v2	Llama 3.3 70B	0.056	0.056	0.011	0.100	90.000	–	–	–	–	–
opensanctions_pairs	Claude Opus 5	0.940	0.906	0.827	0.969	76.249	0.935	0.935	0.874	0.979	0.769
opensanctions_pairs	Claude Sonnet 5	0.730	0.587	0.478	0.704	76.249	0.630	0.636	0.496	0.761	0.769
opensanctions_pairs	DeepSeek-chat	0.760	0.635	0.521	0.753	76.249	0.684	0.691	0.556	0.792	0.769
opensanctions_pairs	DeepSeek-R1	0.850	0.765	0.655	0.860	76.249	0.819	0.819	0.716	0.905	0.769
opensanctions_pairs	Gemini 3.8 Flash	0.860	0.780	0.669	0.881	76.249	0.833	0.833	0.734	0.909	0.769
opensanctions_pairs	GPT-5.4-mini	0.900	0.865	0.782	0.942	76.249	0.891	0.905	0.837	0.963	0.769
opensanctions_pairs	GPT-5.5	0.750	0.608	0.497	0.724	76.249	0.658	0.658	0.520	0.776	0.769
opensanctions_pairs	gpt-oss-120b	0.780	0.666	0.561	0.770	76.249	0.718	0.725	0.594	0.831	0.769
opensanctions_pairs	Grok 4	0.850	0.776	0.677	0.868	76.249	0.824	0.830	0.733	0.914	0.769
opensanctions_pairs	Llama 3.3 70B	0.790	0.682	0.571	0.789	76.249	0.734	0.741	0.618	0.834	0.769
papadakis_dn	Claude Opus 5	0.920	0.955	0.892	0.996	32.601	0.789	0.322	0.130	0.840	0.018

papadakis_dn	Claude Sonnet 5	0.950	0.994	0.987	0.999	32.601	0.857	0.786	0.569	0.947	0.018
papadakis_dn	DeepSeek-chat	0.940	0.993	0.985	0.998	32.601	0.833	0.757	0.520	0.929	0.018
papadakis_dn	DeepSeek-R1	0.940	0.993	0.985	0.998	32.601	0.833	0.757	0.533	0.927	0.018
papadakis_dn	Gemini 3.8 Flash	0.940	0.992	0.984	0.998	32.601	0.824	0.702	0.459	0.889	0.018
papadakis_dn	GPT-5.4-mini	0.950	0.994	0.987	0.999	32.601	0.857	0.786	0.535	0.936	0.018
papadakis_dn	GPT-5.5	0.950	0.994	0.986	0.999	32.601	0.857	0.786	0.573	0.942	0.018
papadakis_dn	gpt-oss-120b	0.930	0.992	0.983	0.997	32.601	0.788	0.682	0.425	0.881	0.018
papadakis_dn	Grok 4	0.950	0.994	0.987	0.999	32.601	0.857	0.786	0.547	0.939	0.018
papadakis_dn	Llama 3.3 70B	0.940	0.975	0.925	0.998	32.601	0.833	0.463	0.180	0.916	0.018
realhitbench	Claude Opus 5	0.700	0.700	0.609	0.791	98.080	–	–	–	–	–
realhitbench	Claude Sonnet 5	0.640	0.640	0.553	0.734	98.080	–	–	–	–	–
realhitbench	DeepSeek-chat	0.620	0.609	0.508	0.701	98.080	–	–	–	–	–
realhitbench	DeepSeek-R1	0.630	0.632	0.534	0.723	98.080	–	–	–	–	–
realhitbench	Gemini 3.8 Flash	0.710	0.708	0.616	0.798	98.080	–	–	–	–	–
realhitbench	GPT-5.4-mini	0.610	0.606	0.506	0.705	98.080	–	–	–	–	–
realhitbench	GPT-5.5	0.690	0.686	0.597	0.776	98.080	–	–	–	–	–
realhitbench	gpt-oss-120b	0.560	0.558	0.460	0.650	98.080	–	–	–	–	–
realhitbench	Grok 4	0.610	0.606	0.507	0.701	98.080	–	–	–	–	–
realhitbench	Llama 3.3 70B	0.530	0.535	0.438	0.632	98.080	–	–	–	–	–
smat	Claude Opus 5	0.900	0.898	0.829	0.951	98.100	–	–	–	–	–
smat	Claude Sonnet 5	0.910	0.907	0.848	0.960	98.100	–	–	–	–	–
smat	DeepSeek-chat	0.880	0.876	0.814	0.936	98.100	–	–	–	–	–
smat	DeepSeek-R1	0.880	0.875	0.808	0.936	98.100	–	–	–	–	–
smat	Gemini 3.8 Flash	0.880	0.876	0.806	0.939	98.100	–	–	–	–	–
smat	GPT-5.4-mini	0.930	0.934	0.881	0.980	98.100	–	–	–	–	–
smat	GPT-5.5	0.910	0.908	0.839	0.958	98.100	–	–	–	–	–
smat	gpt-oss-120b	0.880	0.874	0.806	0.936	98.100	–	–	–	–	–
smat	Grok 4	0.910	0.906	0.847	0.958	98.100	–	–	–	–	–
smat	Llama 3.3 70B	0.680	0.675	0.576	0.765	98.100	–	–	–	–	–
suc	Claude Opus 5	0.980	0.981	0.951	1.000	98.920	–	–	–	–	–
suc	Claude Sonnet 5	0.880	0.874	0.804	0.938	98.920	–	–	–	–	–
suc	DeepSeek-chat	0.820	0.817	0.737	0.891	98.920	–	–	–	–	–
suc	DeepSeek-R1	0.910	0.906	0.844	0.958	98.920	–	–	–	–	–
suc	Gemini 3.8 Flash	0.970	0.971	0.933	1.000	98.920	–	–	–	–	–
suc	GPT-5.4-mini	0.880	0.873	0.802	0.936	98.920	–	–	–	–	–
suc	GPT-5.5	0.920	0.913	0.853	0.966	98.920	–	–	–	–	–
suc	gpt-oss-120b	0.910	0.906	0.842	0.958	98.920	–	–	–	–	–
suc	Grok 4	0.920	0.912	0.849	0.965	98.920	–	–	–	–	–
suc	Llama 3.3 70B	0.670	0.663	0.569	0.755	98.920	–	–	–	–	–
tab_fact	Claude Opus 5	0.660	0.667	0.571	0.760	99.360	0.673	0.696	0.589	0.785	0.500
tab_fact	Claude Sonnet 5	0.740	0.733	0.646	0.818	99.360	0.698	0.710	0.585	0.809	0.500

tab_fact	DeepSeek-chat	0.840	0.826	0.749	0.902	99.360	0.789	0.789	0.676	0.874	0.500
tab_fact	DeepSeek-R1	0.760	0.754	0.669	0.842	99.360	0.721	0.732	0.625	0.825	0.500
tab_fact	Gemini 3.8 Flash	0.860	0.848	0.781	0.924	99.360	0.821	0.821	0.709	0.909	0.500
tab_fact	GPT-5.4-mini	0.820	0.808	0.723	0.883	99.360	0.769	0.772	0.652	0.867	0.500
tab_fact	GPT-5.5	0.840	0.829	0.745	0.902	99.360	0.800	0.803	0.700	0.892	0.500
tab_fact	gpt-oss-120b	0.830	0.815	0.731	0.891	99.360	0.773	0.773	0.656	0.880	0.500
tab_fact	Grok 4	0.850	0.839	0.756	0.912	99.360	0.810	0.812	0.700	0.897	0.500
tab_fact	Llama 3.3 70B	0.780	0.769	0.690	0.849	99.360	0.725	0.732	0.605	0.835	0.500
tabis	Claude Opus 5	0.960	0.964	0.915	0.994	77.069	–	–	–	–	–
tabis	Claude Sonnet 5	0.920	0.917	0.853	0.970	77.069	–	–	–	–	–
tabis	DeepSeek-chat	0.770	0.710	0.609	0.817	77.069	–	–	–	–	–
tabis	DeepSeek-R1	0.970	0.982	0.958	1.000	77.069	–	–	–	–	–
tabis	Gemini 3.8 Flash	0.950	0.970	0.938	0.994	77.069	–	–	–	–	–
tabis	GPT-5.4-mini	0.900	0.905	0.831	0.960	77.069	–	–	–	–	–
tabis	GPT-5.5	0.960	0.976	0.947	0.994	77.069	–	–	–	–	–
tabis	gpt-oss-120b	0.920	0.951	0.914	0.982	77.069	–	–	–	–	–
tabis	Grok 4	0.980	0.988	0.970	1.000	77.069	–	–	–	–	–
tabis	Llama 3.3 70B	0.820	0.821	0.735	0.908	77.069	–	–	–	–	–
tablebench	Claude Opus 5	0.830	0.831	0.749	0.907	89.187	–	–	–	–	–
tablebench	Claude Sonnet 5	0.870	0.867	0.796	0.931	89.187	–	–	–	–	–
tablebench	DeepSeek-chat	0.810	0.820	0.736	0.894	89.187	–	–	–	–	–
tablebench	DeepSeek-R1	0.760	0.773	0.688	0.852	89.187	–	–	–	–	–
tablebench	Gemini 3.8 Flash	0.830	0.829	0.754	0.906	89.187	–	–	–	–	–
tablebench	GPT-5.4-mini	0.830	0.821	0.739	0.898	89.187	–	–	–	–	–
tablebench	GPT-5.5	0.850	0.849	0.769	0.922	89.187	–	–	–	–	–
tablebench	gpt-oss-120b	0.800	0.788	0.700	0.870	89.187	–	–	–	–	–
tablebench	Grok 4	0.820	0.815	0.730	0.897	89.187	–	–	–	–	–
tablebench	Llama 3.3 70B	0.620	0.610	0.510	0.708	89.187	–	–	–	–	–
tableeval	Claude Opus 5	0.740	0.760	0.657	0.860	69.689	–	–	–	–	–
tableeval	Claude Sonnet 5	0.705	0.745	0.639	0.844	69.689	–	–	–	–	–
tableeval	DeepSeek-chat	0.655	0.668	0.552	0.772	69.689	–	–	–	–	–
tableeval	DeepSeek-R1	0.730	0.732	0.632	0.840	69.689	–	–	–	–	–
tableeval	Gemini 3.8 Flash	0.775	0.791	0.693	0.882	69.689	–	–	–	–	–
tableeval	GPT-5.4-mini	0.730	0.744	0.631	0.840	69.689	–	–	–	–	–
tableeval	GPT-5.5	0.760	0.777	0.666	0.870	69.689	–	–	–	–	–
tableeval	gpt-oss-120b	0.760	0.762	0.660	0.863	69.689	–	–	–	–	–
tableeval	Grok 4	0.780	0.769	0.655	0.872	69.689	–	–	–	–	–
tableeval	Llama 3.3 70B	0.570	0.561	0.444	0.680	69.689	–	–	–	–	–
tat_qa	Claude Opus 5	0.810	0.771	0.666	0.868	60.725	–	–	–	–	–
tat_qa	Claude Sonnet 5	0.750	0.733	0.616	0.840	60.725	–	–	–	–	–
tat_qa	DeepSeek-chat	0.740	0.692	0.579	0.813	60.725	–	–	–	–	–

tat_qa	DeepSeek-R1	0.750	0.700	0.586	0.807	60.725	–	–	–	–	–
tat_qa	Gemini 3.8 Flash	0.740	0.709	0.591	0.812	60.725	–	–	–	–	–
tat_qa	GPT-5.4-mini	0.770	0.718	0.591	0.819	60.725	–	–	–	–	–
tat_qa	GPT-5.5	0.730	0.708	0.596	0.810	60.725	–	–	–	–	–
tat_qa	gpt-oss-120b	0.730	0.663	0.543	0.781	60.725	–	–	–	–	–
tat_qa	Grok 4	0.770	0.705	0.580	0.817	60.725	–	–	–	–	–
tat_qa	Llama 3.3 70B	0.720	0.662	0.534	0.776	60.725	–	–	–	–	–
tpcdi_cells	Claude Opus 5	0.350	0.380	0.290	0.477	95.091	–	–	–	–	–
tpcdi_cells	Claude Sonnet 5	0.350	0.380	0.281	0.488	95.091	–	–	–	–	–
tpcdi_cells	DeepSeek-chat	0.350	0.380	0.283	0.480	95.091	–	–	–	–	–
tpcdi_cells	DeepSeek-R1	0.350	0.380	0.284	0.477	95.091	–	–	–	–	–
tpcdi_cells	Gemini 3.8 Flash	0.350	0.380	0.281	0.479	95.091	–	–	–	–	–
tpcdi_cells	GPT-5.4-mini	0.340	0.368	0.262	0.456	95.091	–	–	–	–	–
tpcdi_cells	GPT-5.5	0.350	0.380	0.279	0.474	95.091	–	–	–	–	–
tpcdi_cells	gpt-oss-120b	0.330	0.367	0.264	0.472	95.091	–	–	–	–	–
tpcdi_cells	Grok 4	0.330	0.367	0.274	0.467	95.091	–	–	–	–	–
tpcdi_cells	Llama 3.3 70B	0.330	0.367	0.268	0.461	95.091	–	–	–	–	–
valentine	Claude Opus 5	1.000	1.000	1.000	1.000	81.736	–	–	–	–	–
valentine	Claude Sonnet 5	0.990	0.983	0.935	1.000	81.736	–	–	–	–	–
valentine	DeepSeek-chat	0.970	0.951	0.897	1.000	81.736	–	–	–	–	–
valentine	DeepSeek-R1	0.990	0.986	0.956	1.000	81.736	–	–	–	–	–
valentine	Gemini 3.8 Flash	0.970	0.988	0.970	1.000	81.736	–	–	–	–	–
valentine	GPT-5.4-mini	0.980	0.969	0.922	1.000	81.736	–	–	–	–	–
valentine	GPT-5.5	1.000	1.000	1.000	1.000	81.736	–	–	–	–	–
valentine	gpt-oss-120b	0.970	0.955	0.894	1.000	81.736	–	–	–	–	–
valentine	Grok 4	0.960	0.934	0.867	0.986	81.736	–	–	–	–	–
valentine	Llama 3.3 70B	0.960	0.941	0.875	0.990	81.736	–	–	–	–	–
walmart_amazon	Claude Opus 5	0.980	0.982	0.948	1.000	70.807	0.976	0.912	0.767	1.000	0.094
walmart_amazon	Claude Sonnet 5	0.950	0.962	0.914	0.997	70.807	0.938	0.823	0.645	0.980	0.094
walmart_amazon	DeepSeek-chat	0.950	0.962	0.912	0.998	70.807	0.938	0.823	0.657	0.977	0.094
walmart_amazon	DeepSeek-R1	0.960	0.978	0.939	0.998	70.807	0.950	0.887	0.737	0.989	0.094
walmart_amazon	Gemini 3.8 Flash	0.990	0.998	0.993	1.000	70.807	0.988	0.988	0.959	1.000	0.094
walmart_amazon	GPT-5.4-mini	0.950	0.975	0.938	0.998	70.807	0.937	0.874	0.722	0.987	0.094
walmart_amazon	GPT-5.5	0.970	0.967	0.917	1.000	70.807	0.964	0.848	0.671	1.000	0.094
walmart_amazon	gpt-oss-120b	0.940	0.973	0.933	0.995	70.807	0.923	0.860	0.698	0.974	0.094
walmart_amazon	Grok 4	0.960	0.978	0.942	0.998	70.807	0.950	0.887	0.720	0.989	0.094
walmart_amazon	Llama 3.3 70B	0.930	0.971	0.932	0.994	70.807	0.909	0.847	0.681	0.965	0.094
wdc_lspc	Claude Opus 5	0.990	0.996	0.987	1.000	75.082	0.991	0.992	0.973	1.000	0.264
wdc_lspc	Claude Sonnet 5	0.980	0.992	0.978	1.000	75.082	0.981	0.984	0.958	1.000	0.264
wdc_lspc	DeepSeek-chat	0.960	0.940	0.868	0.994	75.082	0.964	0.895	0.777	0.990	0.264
wdc_lspc	DeepSeek-R1	1.000	1.000	1.000	1.000	75.082	1.000	1.000	1.000	1.000	0.264
wdc_lspc	Gemini 3.8 Flash	0.980	0.992	0.979	1.000	75.082	0.981	0.984	0.956	1.000	0.264

wdc_lspc	GPT-5.4-mini	0.970	0.986	0.967	1.000	75.082	0.971	0.973	0.937	1.000	0.264
wdc_lspc	GPT-5.5	1.000	1.000	1.000	1.000	75.082	1.000	1.000	1.000	1.000	0.264
wdc_lspc	gpt-oss-120b	0.980	0.990	0.974	1.000	75.082	0.981	0.981	0.951	1.000	0.264
wdc_lspc	Grok 4	1.000	1.000	1.000	1.000	75.082	1.000	1.000	1.000	1.000	0.264
wdc_lspc	Llama 3.3 70B	0.960	0.940	0.872	0.994	75.082	0.964	0.895	0.781	0.990	0.264
wdc_products	Claude Opus 5	0.980	0.990	0.974	1.000	77.046	0.981	0.982	0.953	1.000	0.280
wdc_products	Claude Sonnet 5	0.970	0.986	0.967	1.000	77.046	0.971	0.974	0.936	1.000	0.280
wdc_products	DeepSeek-chat	0.970	0.977	0.944	1.000	77.046	0.972	0.958	0.895	1.000	0.280
wdc_products	DeepSeek-R1	0.980	0.990	0.974	1.000	77.046	0.981	0.982	0.953	1.000	0.280
wdc_products	Gemini 3.8 Flash	0.990	0.995	0.983	1.000	77.046	0.991	0.991	0.968	1.000	0.280
wdc_products	GPT-5.4-mini	0.970	0.986	0.966	1.000	77.046	0.971	0.974	0.940	1.000	0.280
wdc_products	GPT-5.5	0.980	0.979	0.940	1.000	77.046	0.981	0.964	0.902	1.000	0.280
wdc_products	gpt-oss-120b	0.980	0.979	0.945	1.000	77.046	0.981	0.964	0.898	1.000	0.280
wdc_products	Grok 4	0.990	0.995	0.983	1.000	77.046	0.991	0.991	0.969	1.000	0.280
wdc_products	Llama 3.3 70B	0.970	0.977	0.940	1.000	77.046	0.972	0.958	0.901	1.000	0.280
wikitablequestions	Claude Opus 5	0.840	0.838	0.760	0.910	97.686	–	–	–	–	–
wikitablequestions	Claude Sonnet 5	0.830	0.829	0.750	0.900	97.686	–	–	–	–	–
wikitablequestions	DeepSeek-chat	0.800	0.798	0.714	0.872	97.686	–	–	–	–	–
wikitablequestions	DeepSeek-R1	0.800	0.797	0.722	0.877	97.686	–	–	–	–	–
wikitablequestions	Gemini 3.8 Flash	0.840	0.839	0.762	0.907	97.686	–	–	–	–	–
wikitablequestions	GPT-5.4-mini	0.810	0.807	0.727	0.879	97.686	–	–	–	–	–
wikitablequestions	GPT-5.5	0.820	0.816	0.737	0.888	97.686	–	–	–	–	–
wikitablequestions	gpt-oss-120b	0.770	0.766	0.684	0.846	97.686	–	–	–	–	–
wikitablequestions	Grok 4	0.810	0.806	0.725	0.884	97.686	–	–	–	–	–
wikitablequestions	Llama 3.3 70B	0.670	0.667	0.583	0.764	97.686	–	–	–	–	–

Table 15: Official-split estimates under the headline rule averaged over the ten models.

dataset	suite_mean	official_ estimate	suite_f1_ pos	official_precision_ pos	official_recall_ pos	official_f1_ pos	official_positive_ share
abt_buy	0.970	0.986	0.968	0.926	0.949	0.936	0.108
alaska_camera	0.958	0.958	0.955	1.000	0.921	0.955	0.500
alaska_schema	0.897	0.897	–	–	–	–	–
amazon_google	0.936	0.932	0.929	0.611	0.942	0.739	0.102
bird	0.126	0.157	–	–	–	–	–
dblp_acm	0.986	0.991	0.985	0.979	0.978	0.976	0.180
dblp_scholar	0.949	0.966	0.948	0.900	0.924	0.910	0.186
fetaqa	0.468	0.468	–	–	–	–	–
fintagging	0.794	0.789	–	–	–	–	–
hitab	0.743	0.759	–	–	–	–	–
machamp	0.887	0.890	0.892	0.644	0.853	0.733	0.175
magneto_gdc	0.908	0.911	–	–	–	–	–
mmtu	0.879	0.834	–	–	–	–	–
officeqa	0.077	0.077	–	–	–	–	–
officeqa_pro_v2	0.082	0.082	–	–	–	–	–
opensanctions_ pairs	0.821	0.727	0.773	0.993	0.649	0.777	0.769
papadakis_dn	0.941	0.988	0.833	0.550	0.963	0.683	0.018
realhitbench	0.630	0.628	–	–	–	–	–
smat	0.876	0.873	–	–	–	–	–
suc	0.886	0.882	–	–	–	–	–
tab_fact	0.798	0.789	0.758	0.896	0.674	0.764	0.500
tabis	0.915	0.918	–	–	–	–	–
tablebench	0.802	0.800	–	–	–	–	–
tableeval	0.720	0.731	–	–	–	–	–
tat_qa	0.751	0.706	–	–	–	–	–
tpcdi_cells	0.343	0.375	–	–	–	–	–
valentine	0.979	0.971	–	–	–	–	–
walmart_amazon	0.958	0.975	0.947	0.832	0.927	0.875	0.094
wdc_lspc	0.982	0.983	0.983	0.965	0.979	0.971	0.264
wdc_products	0.978	0.985	0.979	0.980	0.968	0.974	0.280
wikitablequestions	0.799	0.796	–	–	–	–	–

Table 16: Cross-dataset pairwise tests under rule exact match: Friedman $\chi^2 = 82.5$ ($p = 5.3\text{e-}14$), Iman–Davenport $F = 12.6$, Nemenyi CD = 2.43, Kendall's $W = 0.279$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.3387	21	7	3	0.0266	0.5590	no	no
Claude Opus 5	DeepSeek-chat	-2.6290	22	6	3	0.0049	0.1564	yes	no
Claude Opus 5	DeepSeek-R1	-1.9032	18	6	7	0.0116	0.3133	no	no
Claude Opus 5	Gemini 3.8 Flash	-1.0161	19	8	4	0.0504	0.9126	no	no
Claude Opus 5	GPT-5.4-mini	-2.8065	25	5	1	0.0057	0.1703	yes	no
Claude Opus 5	GPT-5.5	-0.3065	13	9	9	0.3064	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.1935	26	3	2	0.0004	0.0127	yes	yes
Claude Opus 5	Grok 4	-1.7903	18	9	4	0.0559	0.9126	no	no
Claude Opus 5	Llama 3.3 70B	-4.9839	26	5	0	0.0001	0.0049	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.2903	19	5	7	0.0161	0.4034	no	no
Claude Sonnet 5	DeepSeek-R1	-0.5645	14	13	4	0.6305	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.3226	11	14	6	0.3217	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.4677	19	7	5	0.1052	1.0000	no	no
Claude Sonnet 5	GPT-5.5	1.0323	7	15	9	0.0397	0.7936	no	no
Claude Sonnet 5	gpt-oss-120b	-2.8548	22	7	2	0.0094	0.2627	yes	no
Claude Sonnet 5	Grok 4	-0.4516	18	9	4	0.4924	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.6452	25	4	2	0.0001	0.0038	yes	yes
DeepSeek-chat	DeepSeek-R1	0.7258	11	13	7	0.6016	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	1.6129	6	18	7	0.0166	0.4034	no	no
DeepSeek-chat	GPT-5.4-mini	-0.1774	12	14	5	0.4859	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.3226	6	19	6	0.0052	0.1604	no	no
DeepSeek-chat	gpt-oss-120b	-1.5645	19	9	3	0.0827	1.0000	no	no
DeepSeek-chat	Grok 4	0.8387	12	15	4	0.2936	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.3548	22	3	6	0.0001	0.0053	no	yes
DeepSeek-R1	Gemini 3.8 Flash	0.8871	10	17	4	0.0897	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-0.9032	17	12	2	0.9373	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.5968	7	19	5	0.0203	0.4672	no	no
DeepSeek-R1	gpt-oss-120b	-2.2903	21	5	5	0.0038	0.1258	no	no
DeepSeek-R1	Grok 4	0.1129	13	14	4	0.7011	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-3.0806	25	3	3	0.0000	0.0021	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-1.7903	21	9	1	0.0498	0.9126	no	no
Gemini 3.8 Flash	GPT-5.5	0.7097	10	15	6	0.5755	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.1774	24	2	5	0.0001	0.0033	yes	yes
Gemini 3.8 Flash	Grok 4	-0.7742	15	10	6	0.2506	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-3.9677	26	4	1	0.0001	0.0038	yes	yes
GPT-5.4-mini	GPT-5.5	2.5000	5	24	2	0.0064	0.1852	yes	no
GPT-5.4-mini	gpt-oss-120b	-1.3871	20	10	1	0.0134	0.3494	no	no
GPT-5.4-mini	Grok 4	1.0161	9	16	6	0.3613	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.1774	23	4	4	0.0001	0.0044	no	yes
GPT-5.5	gpt-oss-120b	-3.8871	26	1	4	0.0000	0.0004	yes	yes
GPT-5.5	Grok 4	-1.4839	16	6	9	0.0480	0.9126	no	no
GPT-5.5	Llama 3.3 70B	-4.6774	27	4	0	0.0000	0.0009	yes	yes
gpt-oss-120b	Grok 4	2.4032	3	24	4	0.0002	0.0054	no	yes
gpt-oss-120b	Llama 3.3 70B	-0.7903	20	8	3	0.0217	0.4779	no	no
Grok 4	Llama 3.3 70B	-3.1935	24	4	3	0.0001	0.0038	yes	yes

Table 17: Cross-dataset pairwise tests under rule native metric: Friedman $\chi^2 = 80.7$ ($p = 1.2\text{e-}13$), Iman–Davenport $F = 12.2$, Nemenyi CD = 2.43, Kendall's $W = 0.274$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.1613	20	7	4	0.0266	0.5323	no	no
Claude Opus 5	DeepSeek-chat	-2.4677	21	7	3	0.0117	0.2935	yes	no
Claude Opus 5	DeepSeek-R1	-1.7581	18	7	6	0.0210	0.4621	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.5484	15	10	6	0.3505	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.4194	24	5	2	0.0060	0.1748	no	no
Claude Opus 5	GPT-5.5	0.0806	12	10	9	0.5954	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-3.9194	26	4	1	0.0005	0.0177	yes	yes
Claude Opus 5	Grok 4	-1.4839	18	10	3	0.0807	1.0000	no	no
Claude Opus 5	Llama 3.3 70B	-4.7097	25	5	1	0.0001	0.0052	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.3065	20	6	5	0.0349	0.6624	no	no
Claude Sonnet 5	DeepSeek-R1	-0.5968	13	13	5	0.7607	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.6129	11	14	6	0.2550	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.2581	20	7	4	0.0756	1.0000	no	no
Claude Sonnet 5	GPT-5.5	1.2419	8	17	6	0.0089	0.2380	no	no
Claude Sonnet 5	gpt-oss-120b	-2.7581	22	7	2	0.0041	0.1303	yes	no
Claude Sonnet 5	Grok 4	-0.3226	16	12	3	0.8215	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.5484	25	4	2	0.0001	0.0037	yes	yes
DeepSeek-chat	DeepSeek-R1	0.7097	11	15	5	0.4531	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	1.9194	5	20	6	0.0088	0.2380	no	no
DeepSeek-chat	GPT-5.4-mini	0.0484	11	14	6	0.4614	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.5484	5	21	5	0.0009	0.0291	yes	yes
DeepSeek-chat	gpt-oss-120b	-1.4516	19	8	4	0.0651	1.0000	no	no
DeepSeek-chat	Grok 4	0.9839	12	16	3	0.2159	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.2419	23	3	5	0.0001	0.0041	no	yes
DeepSeek-R1	Gemini 3.8 Flash	1.2097	10	17	4	0.0521	0.8959	no	no
DeepSeek-R1	GPT-5.4-mini	-0.6613	16	12	3	0.9061	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.8387	6	20	5	0.0072	0.2017	no	no
DeepSeek-R1	gpt-oss-120b	-2.1613	20	6	5	0.0052	0.1608	no	no
DeepSeek-R1	Grok 4	0.2742	11	16	4	0.2884	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-2.9516	24	4	3	0.0001	0.0041	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-1.8710	20	9	2	0.0498	0.8959	no	no
Gemini 3.8 Flash	GPT-5.5	0.6290	12	15	4	0.6871	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.3710	25	2	4	0.0000	0.0021	yes	yes
Gemini 3.8 Flash	Grok 4	-0.9355	18	11	2	0.1381	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.1613	26	4	1	0.0001	0.0035	yes	yes
GPT-5.4-mini	GPT-5.5	2.5000	5	25	1	0.0053	0.1608	yes	no
GPT-5.4-mini	gpt-oss-120b	-1.5000	21	10	0	0.0120	0.2935	no	no
GPT-5.4-mini	Grok 4	0.9355	10	14	7	0.4149	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.2903	24	3	4	0.0001	0.0022	no	yes
GPT-5.5	gpt-oss-120b	-4.0000	25	1	5	0.0000	0.0004	yes	yes
GPT-5.5	Grok 4	-1.5645	18	6	7	0.0239	0.5022	no	no
GPT-5.5	Llama 3.3 70B	-4.7903	27	4	0	0.0000	0.0008	yes	yes
gpt-oss-120b	Grok 4	2.4355	3	24	4	0.0001	0.0042	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.7903	20	8	3	0.0191	0.4386	no	no
Grok 4	Llama 3.3 70B	-3.2258	23	4	4	0.0001	0.0037	yes	yes

Table 18: Fitted parameters with standard errors (native metric). Kemeny cost 329 of 1395.

model	BT strength	BT SE vs ref	BT bootstrap SD	Rasch ability	Rasch SE	Rasch bootstrap SD	Copeland	Kemeny position
Gemini 3.8 Flash	0.063	0.016	0.005	0.736	0.104	0.078	6.500	3
GPT-5.5	0.060	0.016	0.005	0.693	0.103	0.075	7.500	2
Claude Opus 5	0.045	0.000	0.005	0.651	0.103	0.086	8.500	1
Grok 4	0.031	0.016	0.005	0.345	0.099	0.067	4.500	5
Claude Sonnet 5	0.012	0.016	0.005	0.125	0.097	0.067	4.500	4
DeepSeek-R1	0.012	0.016	0.005	0.070	0.096	0.070	3.500	6
DeepSeek-chat	0.002	0.016	0.005	-0.155	0.094	0.067	1.500	8
GPT-5.4-mini	-0.011	0.016	0.005	-0.084	0.094	0.066	2.500	7
gpt-oss-120b	-0.053	0.016	0.006	-0.608	0.090	0.069	0.500	9
Llama 3.3 70B	-0.161	0.016	0.007	-1.775	0.085	0.083	-0.500	10

Table 19: Mean score per dataset and model under the headline rule (native metric).

dataset	Gemini 3.8 Flash	GPT-5.5	Claude Opus 5	Grok 4	Claude Sonnet 5	DeepSeek-R1	DeepSeek-chat	GPT-5.4-mini	gpt-oss-120b	Llama 3.3 70B
abt_buy	0.980	1.000	1.000	0.960	0.970	0.970	0.960	0.960	0.940	0.960
alaska_camera	0.890	1.000	0.820	0.940	0.980	0.990	1.000	0.970	1.000	0.990
alaska_schema	0.892	0.903	0.925	0.903	0.914	0.925	0.925	0.946	0.903	0.731
amazon_google	0.930	0.940	0.940	0.940	0.960	0.950	0.960	0.930	0.860	0.950
bird	0.210	0.140	0.190	0.080	0.160	0.130	0.120	0.130	0.050	0.050
dblp_acm	1.000	1.000	1.000	1.000	1.000	0.990	1.000	0.990	0.910	0.970
dblp_scholar	0.930	0.960	0.950	0.950	0.970	0.950	0.960	0.940	0.910	0.970
fetaqa	0.557	0.552	0.483	0.536	0.451	0.504	0.540	0.368	0.406	0.287
fintagging	0.820	0.820	0.840	0.810	0.820	0.760	0.750	0.810	0.760	0.750
hitab	0.720	0.730	0.760	0.710	0.740	0.750	0.790	0.760	0.710	0.760
machamp	0.900	0.900	0.900	0.890	0.900	0.900	0.890	0.890	0.830	0.870
magneto_gdc	0.950	0.960	0.960	0.920	0.940	0.960	0.930	0.890	0.950	0.620
mmtu	0.900	0.930	0.900	0.910	0.900	0.830	0.890	0.890	0.830	0.810
officeqa	0.160	0.110	0.150	0.090	0.080	0.040	0.050	0.020	0.050	0.020
officeqa_pro_v2	0.089	0.144	0.167	0.056	0.100	0.022	0.067	0.067	0.056	0.056
opensanctions_pairs	0.860	0.750	0.940	0.850	0.730	0.850	0.760	0.900	0.780	0.790
papadakis_dn	0.940	0.950	0.920	0.950	0.950	0.940	0.940	0.950	0.930	0.940
realhitbench	0.710	0.690	0.700	0.610	0.640	0.630	0.620	0.610	0.560	0.530
smat	0.880	0.910	0.900	0.910	0.910	0.880	0.880	0.930	0.880	0.680
suc	0.970	0.920	0.980	0.920	0.880	0.910	0.820	0.880	0.910	0.670
tab_fact	0.860	0.840	0.660	0.850	0.740	0.760	0.840	0.820	0.830	0.780
tabis	0.950	0.960	0.960	0.980	0.920	0.970	0.770	0.900	0.920	0.820
tablebench	0.830	0.850	0.830	0.820	0.870	0.760	0.810	0.830	0.800	0.620
tableeval	0.775	0.760	0.740	0.780	0.705	0.730	0.655	0.730	0.760	0.570
tat_qa	0.740	0.730	0.810	0.770	0.750	0.750	0.740	0.770	0.730	0.720
tpcdi_cells	0.350	0.350	0.350	0.330	0.350	0.350	0.350	0.340	0.330	0.330
valentine	0.970	1.000	1.000	0.960	0.990	0.990	0.970	0.980	0.970	0.960
walmart_amazon	0.990	0.970	0.980	0.960	0.950	0.960	0.950	0.950	0.940	0.930
wdc_lspc	0.980	1.000	0.990	1.000	0.980	1.000	0.960	0.970	0.980	0.960
wdc_products	0.990	0.980	0.980	0.990	0.970	0.980	0.970	0.970	0.980	0.970
wikitablequestions	0.840	0.820	0.840	0.810	0.830	0.800	0.800	0.810	0.770	0.670

Table 20: Bradley–Terry rank (Borda rank in brackets) under each reported rule, Friedman p and Kendall's W among datasets.

rule	Gemini 3.8 Flash	GPT-5.5	Claude Opus 5	Grok 4	Claude Sonnet 5	DeepSeek-R1	DeepSeek-chat	GPT-5.4-mini	gpt-oss-120b	Llama 3.3 70B	Friedman p	W
native metric	1 (3)	2 (1)	3 (2)	4 (5)	5 (4)	6 (6)	7 (8)	8 (7)	9 (9)	10 (10)	1.2e-13	0.27
exact match	3 (3)	2 (2)	1 (1)	4 (5)	5 (4)	7 (6)	8 (7)	6 (8)	9 (9)	10 (10)	5.3e-14	0.28

Table 21: Kendall τ between the Bradley–Terry rankings of the rules.

rule	native metric	exact match
native metric	1.000	0.778
exact match	0.778	1.000

Table 22: Objective J of each view alone.

view	J alone
mean raw	0.930
baseline-normalised mean	0.886
mean z	0.928
mean win rate	0.931
Borda	0.932
Kemeny score	0.926
Bradley–Terry	0.891
Rasch ability	0.904

Table 23: Split-half honesty check, every split.

split	selected	J_A_selected	J_B_selected	J_B_pure_bt	J_B_uniform	J_B_full_best
0	subset_00010001	0.947	0.853	0.880	0.931	0.905
1	subset_00011010	0.908	0.886	0.838	0.908	0.883
2	lattice_038	0.942	0.933	0.916	0.930	0.944
3	subset_00100011	0.883	0.892	0.869	0.928	0.936
4	subset_01100000	0.933	0.861	0.891	0.897	0.917
5	subset_00111110	0.956	0.919	0.800	0.919	0.911
6	subset_00100010	0.956	0.869	0.889	0.919	0.881
7	subset_10000001	0.930	0.892	0.850	0.886	0.903
8	subset_00110111	0.947	0.947	0.844	0.939	0.906
9	subset_10010010	0.928	0.921	0.921	0.916	0.896
10	subset_01010000	0.972	0.914	0.847	0.928	0.928
11	subset_10110111	0.939	0.936	0.874	0.936	0.955
12	lattice_075	0.964	0.911	0.891	0.897	0.903
13	subset_00000101	0.953	0.947	0.925	0.925	0.950
14	subset_00100000	0.928	0.881	0.872	0.917	0.925
15	subset_00001000	0.937	0.899	0.936	0.919	0.894
16	subset_00010010	0.911	0.911	0.919	0.911	0.883
17	subset_10010101	0.939	0.883	0.906	0.878	0.892
18	lattice_041	0.950	0.925	0.889	0.919	0.931
19	subset_10110101	0.886	0.886	0.821	0.928	0.922

Table 24: Split-half summary.

quantity	value
splits	20
inner_resamples	8
J_A_selected_mean	0.935
J_B_selected_mean	0.903
optimism_gap	0.032
J_B_pure_bt_mean	0.879
J_B_uniform_mean	0.917
J_B_full_best_mean	0.913
selected_minus_pure_bt	"mean": 0.02427983120362126, "ci95": [-0.03387181157278865, 0.11109314599900576], "share_selected_ahead": 0.7

Table 25: The fifteen best candidate mixes with their weights and objective terms.

rank	label	n_views	w_mean_raw	w_baseline_norm_mean	w_z_mean	w_mean_win_rate	w_borda	w_kemeny_score	w_bradley_terry	w_irt_ability	stability	transitivity	reference	J
1	subset_10011001	4	0.250	0.000	0.000	0.250	0.250	0.000	0.000	0.250	0.923	0.944	1.000	0.948
2	subset_10011010	4	0.250	0.000	0.000	0.250	0.250	0.000	0.250	0.000	0.923	0.944	1.000	0.947
3	lhs_038	5	0.000	0.200	0.100	0.350	0.100	0.000	0.000	0.250	0.920	0.944	1.000	0.946
4	subset_11111010	6	0.167	0.167	0.167	0.167	0.167	0.000	0.167	0.000	0.920	0.944	1.000	0.946
5	lattice_076	7	0.250	0.050	0.150	0.150	0.200	0.000	0.100	0.100	0.920	0.944	1.000	0.946
6	subset_01011001	4	0.000	0.250	0.000	0.250	0.250	0.000	0.000	0.250	0.920	0.944	1.000	0.946
7	lattice_027	7	0.050	0.200	0.150	0.100	0.350	0.000	0.050	0.100	0.920	0.944	1.000	0.946
8	subset_01111001	5	0.000	0.200	0.200	0.200	0.200	0.000	0.000	0.200	0.919	0.944	1.000	0.946
9	lattice_083	6	0.000	0.150	0.100	0.400	0.050	0.050	0.000	0.250	0.919	0.944	1.000	0.946
12	lhs_095	4	0.150	0.250	0.000	0.450	0.000	0.000	0.150	0.000	0.919	0.944	1.000	0.945
11	lattice_071	7	0.100	0.100	0.150	0.300	0.100	0.000	0.100	0.150	0.919	0.944	1.000	0.945
10	lhs_032	6	0.150	0.200	0.150	0.150	0.300	0.000	0.050	0.000	0.919	0.944	1.000	0.945
13	subset_11011001	5	0.200	0.200	0.000	0.200	0.200	0.000	0.000	0.200	0.918	0.944	1.000	0.945
14	subset_01111010	5	0.000	0.200	0.200	0.200	0.200	0.000	0.200	0.000	0.918	0.944	1.000	0.945
15	subset_10010010	3	0.333	0.000	0.000	0.333	0.000	0.000	0.333	0.000	0.918	0.944	1.000	0.945

Table 26: Value of each view per model (exact-match scores).

model	mean raw	baseline-normalised mean	mean z	mean win rate	Borda	Kemeny score	Bradley–Terry	Rasch ability
Gemini 3.8 Flash	0.763	0.738	0.386	0.620	5.581	7.000	0.037	0.126
GPT-5.5	0.764	0.740	0.526	0.699	6.290	8.000	0.039	0.130
Claude Opus 5	0.768	0.737	0.631	0.733	6.597	9.000	0.047	0.162
Grok 4	0.754	0.730	0.155	0.534	4.806	5.000	0.020	0.062
Claude Sonnet 5	0.751	0.721	0.228	0.584	5.258	6.000	0.013	0.039
DeepSeek-R1	0.748	0.722	0.142	0.522	4.694	4.000	0.008	0.018
DeepSeek-chat	0.737	0.710	-0.092	0.441	3.968	2.000	-0.014	-0.054
GPT-5.4-mini	0.749	0.724	-0.040	0.421	3.790	3.000	0.010	0.028
gpt-oss-120b	0.728	0.690	-0.730	0.267	2.403	1.000	-0.033	-0.116
Llama 3.3 70B	0.680	0.649	-1.205	0.179	1.613	0.000	-0.128	-0.395

Table 27: Value of each view per model (native-metric scores).

model	mean raw	baseline-normalised mean	mean z	mean win rate	Borda	Kemeny score	Bradley–Terry	Rasch ability
Gemini 3.8 Flash	0.792	0.767	0.438	0.643	5.790	7.000	0.063	0.176
GPT-5.5	0.793	0.769	0.570	0.713	6.419	8.000	0.060	0.167
Claude Opus 5	0.792	0.762	0.556	0.704	6.339	9.000	0.045	0.157
Grok 4	0.780	0.756	0.181	0.539	4.855	5.000	0.031	0.086
Claude Sonnet 5	0.776	0.746	0.212	0.575	5.177	6.000	0.012	0.033
DeepSeek-R1	0.772	0.746	0.083	0.509	4.581	4.000	0.012	0.020
DeepSeek-chat	0.763	0.736	-0.122	0.430	3.871	2.000	0.002	-0.035
GPT-5.4-mini	0.771	0.746	0.010	0.436	3.919	3.000	-0.011	-0.018
gpt-oss-120b	0.750	0.712	-0.732	0.269	2.419	1.000	-0.053	-0.148
Llama 3.3 70B	0.701	0.670	-1.197	0.181	1.629	0.000	-0.161	-0.439